Tomita–Takesaki modular structure on truncated SU(2) spectral triples: four-witness Wick-rotation literal identification at finite cutoff

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Abstract

The four-witness Wick-rotation theorem (Hartle–Hawking [13], Sewell [24], Bisognano–Wichmann [2], Unruh [29]) asserts that a state of a quantum field restricted to a region bounded by a bifurcate Killing horizon with surface gravity κ_g is KMS at inverse temperature $\beta = 2\pi/\kappa_g$ with respect to Killing-time evolution. The shared 2π across all four faces is $\mathrm{Vol}(S^1)$, the regularity period of the Euclidean rotation of the observer's causal-domain coordinates. We close this theorem at the operator-system level on the Connes–van Suijlekom [10] truncated Camporesi–Higuchi [4] spectral triple $\mathcal{T}_{n_{\max}}$ via two structurally independent but operator-action conjugate constructions: (i) the geometric BW-α realization $K_\alpha = J_{\text{polar}}$ with integer spectrum on the spinor basis, and (ii) the Tomita–Takesaki BW-γ realization $K_{\text{TT}} = -\log\Delta$ obtained from polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space of a wedge-restricted Gibbs state. Both constructions verify the period closure $\sigma_{2\pi}(O) = O$ *bit-exactly* at every tested $n_{\max} \in \{2, 3, 4, 5\}$ for all four physical witnesses, with maximum periodicity residual scaling as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\text{machine}})$. Cross-witness collapse is bit-exact at the algebra-action level: the density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is itself β-independent under the natural choice $H_{\text{local}} := K_\alpha^W/\beta$, so the six witness instantiations (BW canonical, Hartle–Hawking $M = 1$, Hartle–Hawking $M = 2$, Sewell $M = 1$, Unruh $a = 1$, Unruh $a = 2$) give literally identical modular operators Δ and Hamiltonians K_{TT}. We further establish a flow-conjugacy identity $\sigma_t^{\text{TT}}(O) = \sigma_{-t}^{\alpha}(O)$ bit-exact at general t, identical at the period $t = 2\pi$. The construction depends on a deliberate choice of local Hamiltonian $H_{\text{local}} = K_\alpha^W/\beta$ on the wedge sub-algebra, the operator-system analog of the Wightman BW vacuum. The framework's intrinsic Camporesi–Higuchi Dirac, restricted to the wedge, would *not* produce the bit-exact closure under the same wedge-state prescription, isolating $H_{\text{local}} = K_\alpha^W/\beta$ as a derived structural finding. The L1 closure lifts the four-witness theorem from "structural correspondence" (the published Wick-rotation chain at the metric-functional level) to "literal identification at the operator-system level (Riemannian)" at every finite cutoff. The Lorentzian extension to signature $(3, 1)$ via Bizi–Brouder–Besnard 2018 [3] Krein-space spectral triples is carried out *at finite cutoff* in the present paper (§11); its *continuum* (Lorentzian-propinquity-rate) closure remains the named multi-month follow-up. The result is the metric-spectral-triple analog of the Connes–Rovelli thermal time hypothesis [9] restricted to a hemispheric wedge of the round S^3.

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Keywords: Tomita–Takesaki modular flow, Bisognano–Wichmann theorem, Connes–van Suijlekom truncated operator system, Camporesi–Higuchi Dirac, four-witness Wick rotation, Hartle–Hawking, Sewell, Unruh effect, KMS state, SU(2) spectral triple.

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1 Introduction

A central thread of axiomatic quantum field theory is the *four-witness Wick-rotation theorem*: under suitable analytic hypotheses, a state of a quantum field restricted to a region bounded by a bifurcate Killing horizon with surface gravity κ_g is a KMS state at inverse temperature $\beta = 2\pi/\kappa_g$ with respect to evolution along the relevant Killing vector field. The four physical instantiations are Bisognano–Wichmann [2] for the Rindler wedge of Minkowski spacetime, Hartle–Hawking [13] for the Hartle–Hawking thermal vacuum on the Schwarzschild exterior, Sewell [24] for the abstract KMS-modular reading of the Hartle–Hawking vacuum on globally hyperbolic manifolds with bifurcate Killing horizons, and Unruh [29] for the proper-acceleration parameterisation of the Bisognano–Wichmann setup. The shared 2π across all four faces is $\mathrm{Vol}(S^1)$, the regularity period of the Euclidean rotation of the observer's causal-domain coordinates after Wick rotation $\tau = it$.

The theorem closes naturally in axiomatic Wightman QFT for free fields on globally hyperbolic Lorentzian spacetimes. Its operator-system analog on a finite-dimensional truncation of a spectral triple — i.e., the question whether the modular automorphism associated to a wedge-restricted Gibbs state on a finite-dimensional operator system realises the same $\sigma_{i\cdot 2\pi} =$ identity period closure intrinsically, without recourse to the Wightman axiomatic Wick-rotation chain — is structurally distinct, and has not been settled in the spectral-truncation literature of [10, 19].

The present paper closes this operator-system analog on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ over the round $S^3 = \mathrm{SU}(2)$. We work in the setup of Connes–van Suijlekom [10], with the truncated operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}}$ obtained by orthogonal projection onto the finite-dimensional subspace $\mathcal{H}_{n_{\max}}$ spanned by the lowest Camporesi–Higuchi spinor harmonics, and with the Connes–vS "Toeplitz S^1" propagation invariant $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ verified verbatim in supporting work of the present author (Paper 32 § III). The convergence of $\mathcal{T}_{n_{\max}}$ to the round-S^3 Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance — with explicit rate $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq C_3 \cdot \gamma_{n_{\max}}$ where $\gamma_{n_{\max}} \sim (4/\pi)\log n_{\max}/n_{\max}$ and $C_3 = 1$ — has been established in the present author's Paper 38 [37].

Theorem 1.1 (Main theorem; cf. Section 7)**.** *Let $\mathcal{T}_{n_{\max}}$ denote the truncated Camporesi–Higuchi spectral triple of cutoff $n_{\max}$ over the round $S^3 = \mathrm{SU}(2)$, let P_W denote the hemispheric wedge projector defined by the equatorial reflection $R_{\mathrm{polar}} : m_j \mapsto -m_j$ on the full-Dirac spinor basis, and let $\omega_W^{\mathrm{BW}}(O) = Z^{-1}\mathrm{Tr}_{P_W\mathcal{H}_{n_{\max}}}(e^{-K_\alpha^W}\, O)$ denote the wedge-restricted Gibbs state at the BW local Hamiltonian $H_{\mathrm{local}} = K_\alpha^W/\beta$. Then for every $n_{\max} \geq 1$, both the geometric BW-α modular flow $\sigma_t^\alpha(O) = e^{itK_\alpha^W} O e^{-itK_\alpha^W}$ and the Tomita–Takesaki BW-γ modular flow $\sigma_t^{\mathrm{TT}}(O) = \Delta^{it} O \Delta^{-it}$, with $\Delta = S^*S$ extracted from the polar decomposition $S = J_{\mathrm{TT}}\Delta^{1/2}$ on the GNS Hilbert–Schmidt space, satisfy the period closure*

$$\sigma_{2\pi}^\alpha(O) \;=\; O, \qquad \sigma_{2\pi}^{\mathrm{TT}}(O) \;=\; O \tag{1}$$

bit-exactly for every $O \in P_W\mathcal{O}_{n_{\max}}P_W$, with maximum periodicity residual scaling as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\mathrm{machine}})$. Furthermore, the two flows are operator-action conjugates,

$$\sigma_t^{\mathrm{TT}}(O) \;=\; \sigma_{-t}^\alpha(O) \tag{2}$$

bit-exact at every real t, and the four-witness theorem collapses to a single operator-system construction at the algebra-action level: the density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is β-independent, so the six witness instantiations $(\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2})$ *produce literally identical Δ and K_{TT}.*

The result is not a theorem in the sense of an analytic limit statement; it is a finite-cutoff bit-exact algebraic identity, with the structural reason being that the BW-α geometric generator

$K_\alpha^W = J_{\mathrm{polar}}$ has *integer spectrum* two_m_j on the full-Dirac basis (odd integers in $\{-(2l+1), \ldots, +(2l+1)\}$), so $e^{i \cdot 2\pi \cdot n} = 1$ for any integer n produces the period closure pointwise at every $n_{\max}$ without requiring a Gromov–Hausdorff limit. The closure therefore complements the Paper 38 state-space Gromov–Hausdorff convergence at the operator level: Paper 38 gives a qualitative-rate state-space-GH bound $O(\log n_{\max}/n_{\max})$ on the metric-spectral-triple data; the present paper establishes the modular period closure *at machine precision at every finite cutoff* as a structural property of the spectral content rather than a limit statement.

What this paper does and does not close. This paper closes the principal Sprint L1 falsifier named in Paper 32 § VIII Remark `rem:bisognano_wichmann_reading` of the present author's framework documentation: "compute the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i \cdot 2\pi}$ = identity at finite $n_{\max}$." The verdict is STRONG_IDENTIFICATION via BW-α and UNIFIED_STRONG via BW-γ. The Lorentzian extension to signature $(3,1)$ via the Bizi–Brouder–Besnard 2018 [3] Krein-space spectral triple is carried out *at finite cutoff* in §11 (the operator-system four-witness period closure $\sigma_{2\pi} = \mathrm{id}$ on the Krein wedge); the continuum / Lorentzian-propinquity-rate extension is *not* closed by this paper (the named open target, and Paper 45's degeneracy theorem closes off the natural Lorentzian-propinquity route). Sprint L0 of the present author's framework documentation (`debug/lorentzian_l0_audit_memo.md`, Paper 31 § 8, Paper 34 § V.E) audits the 28 projections of the framework's projection taxonomy and predicts that the M3 sub-mechanism of the master Mellin engine becomes trivially empty at signature $(3,1)$ on chirality-symmetric spectrum, leaving the M1 $\mathrm{Vol}(S^1) = 2\pi$ mechanism (the load-bearing engine for the four-witness theorem) intact under signature change. The L1 closure of the present paper is the Riemannian-side anchor that confirms the operator-system identification is sound at signature $(3,0)$, before lifting signature.

Related and concurrent work. The most directly comparable construction in the published literature is the Connes–Rovelli [9] thermal time hypothesis, which identifies the Tomita modular flow on a covariant quantum field theory as the physical time evolution. The Connes–Rovelli identification is general-covariant and abstract; the present paper specialises it to a finite-dimensional operator system in the Connes–van Suijlekom [10] truncation framework on the round S^3. The continuum side has the long tradition starting from Bisognano–Wichmann [2] and standardised through Casini–Huerta [5] and Casini–Huerta–Myers [6], with the continuum modular Hamiltonian for a hemisphere of S^d in CFT given explicitly as $K_{\mathrm{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, d\Omega$ where ξ is the wedge-preserving Killing field. The present finite-cutoff structural correspondence reproduces the form of this identification on the truncated operator system, with the integer spectrum of the BW-α generator playing the role of the canonical 2π-period of the continuum modular flow. A finite-lattice realisation in critical quantum spin chains was given by Zhang *et al.* [31]; the present construction differs in that the underlying object is a continuum spectral triple on a non-flat compact S^3 background, and the finite-dimensional structure comes from the spectral truncation of D_{CH} rather than from a discretisation of physical space. The existing modular literature on Riemannian backgrounds includes Strohmaier [25] (Reeh–Schlieder on stationary spacetimes) and Verch [30] (generally covariant spin-statistics and modular structure on curved backgrounds); these address the algebraic-QFT side, whereas the present paper anchors the modular structure on the Connes–van Suijlekom truncated operator system. Direct attempts to lift the Wick-rotation chain at the spectral-triple level via twisted-spectral-triple morphisms have been pursued recently by Nieuviarts [21–23], but the twist construction (Definition 2.2 of [22]) is explicitly restricted to even-dimensional manifolds and does not apply to the S^3 case treated here.

Outline. Section 2 recalls the truncated Camporesi–Higuchi spectral triple and the Paper 38 state-space Gromov–Hausdorff convergence result. Section 3 summarises the Sprint L0 Lorentzian-readiness audit, names the Riemannian scope of the present paper, and footnotes the Nieuviarts NO-GO on direct twist-morphism emergence for odd-dim S^3. Section 4 sets up the hemispheric wedge P_W, the KMS state at $\beta = 2\pi$ on the wedge sub-algebra, and the unified surface-gravity normalisation. Section 5 constructs the geometric BW-α modular Hamiltonian $K_\alpha = J_{\text{polar}}$ and proves Theorem 5.4 (Sprint L1 result, STRONG_IDENTIFICATION). Section 6 constructs the Tomita–Takesaki BW-γ modular Hamiltonian $K_{\text{TT}} = -\log\Delta$ via polar decomposition on the GNS Hilbert–Schmidt space, distinguishes J_{TT} ($J^2 = +I$) from the Connes J_{GV} ($J^2 = -I$), and proves Theorem 6.3 (Sprint L1-tighten result, UNIFIED_STRONG). Section 7 proves Theorem 7.1 (the flow conjugacy $\sigma_t^{\text{TT}} = \sigma_{-t}^{\alpha}$) and discusses the BW-aligned local Hamiltonian choice $H_{\text{local}} = K_\alpha^W/\beta$ as a derived structural finding rather than a uniqueness theorem. Section 8 establishes the cross-witness collapse of HH, Sewell, BW, and Unruh into a single operator-system construction. Section 9 places the result in the modular-theory and NCG literature. Section 10 summarises and names open questions. Appendix A documents the computational verification protocol and the supporting module `geovac/modular_hamiltonian.py`. Appendix B cross-references the result to the present author's broader framework (Paper 38 [37] for state-space Gromov–Hausdorff convergence; Paper 40 [39] for the universal $4/\pi$ rate constant; Paper 32 § VIII for the master Mellin engine case-exhaustion theorem and the four-witness theorem's M1 sub-mechanism reading).

2 The truncated Camporesi–Higuchi spectral triple

We work in the framework of metric spectral triples and their spectral truncations established by Connes [8, 10] and developed in the GH-convergence context by Latrémolière [17, 18], Hekkelman [14], Leimbach–van Suijlekom [19], and the present author's Paper 38 [37] for the round $S^3 = \mathrm{SU}(2)$ case.

2.1 The continuum Camporesi–Higuchi triple

Identify the unit round S^3 with the Lie group SU(2) via the standard double cover, equipped with the bi-invariant metric of total volume $\mathrm{Vol}(S^3) = 2\pi^2$. Let Σ denote the spinor bundle and $L^2(S^3, \Sigma)$ the spinor Hilbert space. The *Camporesi–Higuchi Dirac operator* $D_{\text{CH}} : \mathrm{Dom}(D_{\text{CH}}) \subset L^2(S^3, \Sigma) \to L^2(S^3, \Sigma)$ acts diagonally on the spinor harmonic basis $\{\psi_{n,l,m_j,\chi}\}$ via

$$D_{\text{CH}}\, \psi_{n,l,m_j,\chi} \;=\; \chi\,(n + \tfrac{1}{2})\,\psi_{n,l,m_j,\chi}, \tag{3}$$

with $n \in \mathbb{N}_{\geq 1}$, $l \in \{0, 1, \ldots, n-1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \ldots, j\}$, and chirality $\chi \in \{+1, -1\}$ [4]. The degeneracy of the level n on each chirality sector is $n(n+1)$; the total degeneracy including both chiralities is $2n(n+1)$.

We form the spectral triple

$$\mathcal{T}_{S^3} \;=\; \big(C^\infty(S^3),\; L^2(S^3, \Sigma),\; D_{\text{CH}}\big). \tag{4}$$

This is a unital, even, real spectral triple of metric dimension 3 in the sense of Connes [8], with KO-dimension 3 (mod 8). The chirality grading $\gamma_5 = \mathrm{diag}(+1, -1)$ on $\Sigma_+ \oplus \Sigma_-$, the Connes real structure J_{GV} (charge conjugation, $J_{\text{GV}}^2 = -I$), and the standard Lipschitz seminorm $\|[D_{\text{CH}}, M_f]\|_{\text{op}}$ on the algebra $C^\infty(S^3)$ all enter as expected. We use the labelling $n_{\text{Fock}} = n_{\text{CH}} + 1$ throughout, so the lowest non-trivial level is $n = 1$ with $|\lambda_1| = 3/2$.

2.2 Truncations and the operator system

Let $P_{n_{\max}} : L^2(S^3, \Sigma) \to \mathcal{H}_{n_{\max}}$ denote the orthogonal projection onto the finite-dimensional subspace spanned by the spinor harmonics with $1 \leq n \leq n_{\max}$. The truncated Hilbert space has dimension

$$\dim \mathcal{H}_{n_{\max}} \;=\; 2\sum_{n=1}^{n_{\max}} n(n+1) \;=\; \tfrac{2}{3}\, n_{\max}(n_{\max}+1)(n_{\max}+2). \tag{5}$$

Numerically: $\dim \mathcal{H}_2 = 16$, $\dim \mathcal{H}_3 = 40$, $\dim \mathcal{H}_4 = 80$, $\dim \mathcal{H}_5 = 140$. The $n_{\max} = 3$ case is structurally distinguished by the identity $\dim \mathcal{H}_3 = 40 = g_3^{\text{Dirac}} = \Delta^{-1}$, where g_n^{Dirac} is the single-chirality Dirac mode degeneracy on S^3 at level n and Δ^{-1} is the boundary-mass invariant of the present author's Paper 2 [32] fine-structure-constant observation.

The truncated Dirac operator is $D_{n_{\max}} = P_{n_{\max}} D_{\text{CH}} P_{n_{\max}}$, which (since D_{CH} is diagonal in the spinor basis) acts as $\chi\,(n+1/2)$ on each kept basis vector. Following Connes–van Suijlekom [10], the *truncated operator system* is

$$\mathcal{O}_{n_{\max}} \;=\; P_{n_{\max}}\, C^{\infty}(S^3)\, P_{n_{\max}} \;\subset\; B(\mathcal{H}_{n_{\max}}), \tag{6}$$

viewed as a self-adjoint subspace of the matrix algebra $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$. As emphasised in [10, §1], this is *not* a C^*-subalgebra of $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$: the full C^*-algebra it generates is the full matrix algebra (the "C^*-envelope"), but the operator system $\mathcal{O}_{n_{\max}}$ itself encodes finer structure, including the algebra-side analog of the Lipschitz seminorm and the propagation invariant. The truncated metric spectral triple is then $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$.

Paper 32 § III of the present author's framework documentation verifies that the propagation invariant $\text{prop}(\mathcal{O}_{n_{\max}}) = 2$ at $n_{\max} \in \{2, 3, 4\}$, matching Connes–van Suijlekom [10, Prop. 4.2] for the Toeplitz operator system on S^1 verbatim. This is the structural match that establishes $\mathcal{O}_{n_{\max}}$ as the right operator-system host for spectral-triple-internal modular constructions.

2.3 Propinquity convergence to the continuum

The present author's Paper 38 [37] proves that the truncated triples $\mathcal{T}_{n_{\max}}$ converge to $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance with explicit rate

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; C_3 \cdot \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} \;=\; \frac{4 \log n_{\max}}{\pi\, n_{\max}} \;+\; O(1/n_{\max}), \tag{7}$$

with Lipschitz comparison constant $C_3 = 1$ and asymptotic rate constant $4/\pi = \text{Vol}(S^2)/\pi^2$. The constant $4/\pi$ identifies as the Hopf-base measure factor in the standard $\text{SU}(2)/U(1)$ Haar normalisation, the M1 sub-mechanism signature of the master Mellin engine on S^3 (Paper 32 § VIII case-exhaustion theorem; generalised to all compact connected Lie groups in Paper 40 [39]). The five lemmas L1' / L2 / L3 / L4 / L5 of Paper 38 — chirality-doubled operator system substrate, central spectral Fejér kernel on SU(2), Lipschitz comparison bound, Berezin reconstruction, Latrémolière propinquity assembly — supply the analytic infrastructure that the present paper builds on.

The present paper does not require the GH-convergence machinery in any load-bearing way: the period closure proved below is a finite-cutoff bit-exact identity, not a limit statement. The state-space Gromov–Hausdorff convergence (7) enters only as a *consistency* cross-check (Section 6), confirming that the modular construction inherits the M1 transcendental signature of the metric-side state-space GH convergence.

3 Lorentzian-readiness audit and Riemannian scope

The setting of the present paper is signature $(3,0)$: the round S^3 Camporesi–Higuchi spectral triple is Riemannian, with KO-dimension 3 (mod 8). The four-witness Wick-rotation theorem inhabits a hybrid status: each of the four physical witnesses (Hartle–Hawking, Sewell, Bisognano–Wichmann, Unruh) is naturally a *Lorentzian* statement at signature $(3,1)$ in the Wightman axiomatic framework, but the shared $\beta = 2\pi/\kappa_g$ structural identity that unifies the four faces lives entirely on the Euclidean side via Wick rotation. The Wick-rotation chain itself [Hartle–Hawking 1976 → Sewell 1982 → Bisognano–Wichmann 1976 → Unruh 1976] operates at the metric-functional level: it converts a Euclidean heat-kernel-style spectral computation into its Lorentzian modular-flow interpretation without re-deriving the underlying spectral object in Krein signature.

3.1 The 28-projection Lorentzian transfer audit

The present author's Sprint L0 audit (`debug/lorentzian_l0_audit_memo.md`, Paper 31 § 8, Paper 34 § V.E) classifies the 28 projections of the framework's projection taxonomy into four buckets:

(T) *TRANSFERS_FREELY:* signature-blind, lives on the variable/dimension axes of the projection taxonomy only, with no Riemannian volume form, no Mellin transform, and no Latrémolière propinquity. Holds at $(3,0)$ and $(3,1)$ identically.

(W) *WICK_MAP_FREE:* reduces to the M1 sub-mechanism of the master Mellin engine via the four-witness Wick-rotation theorem at the metric-functional level.

(E) *EUCLIDEAN_SPECIFIC:* requires either (i) a Krein-space spectral triple at $(3,1)$ per Bizi–Brouder–Besnard 2018 [3], (ii) a Lorentzian-native heat-trace formalism, or (iii) a Lorentzian analog of the Latrémolière propinquity (which does not yet exist in the published literature).

(M) *MIXED:* parts transfer freely, parts require extension.

The headline distribution at May 2026 is $17/4/5/2 = 61\%/14\%/18\%/7\%$: three-quarters of the projection dictionary inherits a Lorentzian reading at no further structural cost at the metric-functional level. The Wick-rotation projection itself sits in the (W) bucket and carries the M1 transcendental signature $2\pi\cdot\mathbb{Q}$ via $\mathrm{Vol}(S^1)$. The five EUCLIDEAN_SPECIFIC projections (Fock conformal, Hopf bundle, stereographic, spectral action, Camporesi–Higuchi spinor lift) all require a full Krein-space lift to extend cleanly to signature $(3,1)$.

The present paper does not address signature $(3,1)$. It operates entirely on the Riemannian Camporesi–Higuchi triple at signature $(3,0)$ and closes the modular-period identity inside that framework. The Sprint L0 audit identified the operator-system literal-identification of the Wick-rotation projection at signature $(3,0)$ as the principal Sprint L1 falsifier ahead of the multi-month $(3,1)$ extension; the present paper closes exactly that falsifier.

3.2 Footnote on the Nieuviarts twist morphism

A natural shortcut to the signature change would be the twisted-spectral-triple morphism of Nieuviarts [21–23], which constructs a pseudo-Riemannian spectral triple from an almost-commutative twisted Riemannian spectral triple, presented as an alternative to Wick rotation. The construction is mechanistically appealing but does *not* apply to the present S^3 case: Definition 2.2 of [22] requires a $\mathbb{Z}_2$ grading Γ that commutes with the algebra and anti-commutes with the Dirac, which exists only for *even* KO-dimension spectral triples. The S^3 Camporesi–Higuchi triple has KO-dimension $3 \pmod 8$ (odd); the restriction is structural rather than editorial (the author of [22] explicitly notes: “the fact that no such procedure for twisting odd-dimensional manifold's spectral triple has been found will be one of the justifications to focus on the study of even-dimensional manifolds”). The Lorentzian-extension path for the present construction therefore runs through the direct Bizi–Brouder–Besnard 2018 Krein-space lift, not through a twist morphism; this is the named Sprint L2 follow-up of the present author's framework documentation. A dedicated NO-GO analysis of the Nieuviarts route on $S^3 = \mathrm{SU}(2)$ is recorded in `debug/nieuviarts_scoping_memo.md` of the present author's internal materials.

4 The Bisognano–Wichmann modular structure at finite cutoff

We set up the operator-system-level analog of the Bisognano–Wichmann construction on the truncated Camporesi–Higuchi spectral triple. Three ingredients are required: (i) the hemispheric wedge projector P_W, (ii) the wedge-restricted Gibbs state ω_W at inverse temperature β, (iii) the local Hamiltonian H_{local} generating the wedge-modular evolution.

4.1 The hemispheric wedge projector

Definition 4.1 (Hemispheric wedge on $\mathcal{H}_{n_{\max}}$)**.** The *Hopf-axis equatorial reflection* on $\mathcal{H}_{n_{\max}}$ is the involution

$$R_{\text{polar}} : \psi_{n,l,m_j,\chi} \longmapsto \psi_{n,l,-m_j,\chi}, \tag{8}$$

acting as $m_j \mapsto -m_j$ on the spinor basis. The *hemispheric wedge projector* is

$$P_W \;:=\; \tfrac{1}{2}(I + R_{\text{polar}}) \;\in\; B(\mathcal{H}_{n_{\max}}). \tag{9}$$

Proposition 4.2 (Wedge projector properties)**.** *The hemispheric wedge projector P_W satisfies:*

(a) $P_W^2 = P_W$ *(idempotent);*

(b) $P_W^* = P_W$ *(Hermitian);*

(c) $\mathrm{Tr}(P_W) = \dim \mathcal{H}_{n_{\max}}/2$ *(half-dimensional projection).*

Equivalently, $\mathcal{H}_{n_{\max}} = \mathcal{H}_W \oplus \mathcal{H}_{W'}$ *where* $\mathcal{H}_W = P_W \mathcal{H}_{n_{\max}}$ *has dimension* $\dim_W = \dim \mathcal{H}_{n_{\max}}/2$*, i.e.,* $\dim_W = 8, 20, 40, 70$ *at* $n_{\max} = 2, 3, 4, 5$.

Proof. (a) and (b) follow from $R_{\text{polar}}^2 = I$ and $R_{\text{polar}}^* = R_{\text{polar}}$, the latter holding because R_{polar} is a permutation matrix. For (c): on the full-Dirac spinor basis the values two_m_j are *odd integers*, so the involution $m_j \mapsto -m_j$ has no fixed points, i.e., R_{polar} has spectrum $\{+1, -1\}$ each with multiplicity $\dim \mathcal{H}_{n_{\max}}/2$. □

The hemispheric wedge $\mathcal{H}_W$ is the operator-system analog of the half-S^3 algebra $C^\infty(H_+)$ where $H_+ = \{\omega \in S^3 : \omega \cdot \hat{n} > 0\}$ is the open hemisphere of S^3 aligned with a chosen polar axis $\hat{n}$. The choice of axis is fixed canonically by alignment with the Hopf-base direction (see [34, §V.3] for the role of the Hopf-base axis in the present author's framework documentation, and Section 6 below for the role of the integer spectrum of two_m_j on this axis).

Remark 4.3 (On the choice of wedge)**.** Three candidates for the wedge sub-system were named in the architectural scoping (`debug/l1_modular_hamiltonian_architecture_memo.md` §3): (W1) the hemispheric wedge of Definition 4.1; (W2) the κ-preserving block of Paper 29's Dirac graph Rule A; (W3) the pendant-vertex-free subgraph of the scalar Fock graph. Candidate W1 is the cleanest Riemannian-side analog of the Wightman Rindler wedge (a region bounded by a horizon of codimension 1 in the underlying manifold), and it carries a natural Killing-preserving structure (the rotation around the Hopf-base axis preserves the wedge boundary). Candidates W2 and W3 are graph-topological projections rather than spatial-region restrictions, and their modular structure either degenerates (W2) or has no clear "localised in a half-region" interpretation (W3). The present paper proceeds with W1 as the canonical choice. The dependence of the closure on the wedge choice is part of the open scope (Section 10).

4.2 The wedge-restricted Gibbs state

For a self-adjoint operator H on $\mathcal{H}_W$ and a positive inverse temperature $\beta > 0$, the Gibbs state $\omega_{\beta,H} : B(\mathcal{H}_W) \to \mathbb{C}$ is

$$\omega_{\beta,H}(O) \;=\; \frac{\mathrm{Tr}_{\mathcal{H}_W}\big(e^{-\beta H}\, O\big)}{\mathrm{Tr}_{\mathcal{H}_W}\big(e^{-\beta H}\big)}, \qquad O \in B(\mathcal{H}_W). \tag{10}$$

Restricted to the wedge sub-operator-system $\mathcal{O}_{W,n_{\max}} := P_W \mathcal{O}_{n_{\max}} P_W$, this is the natural Hilbert-space-internal analog of the Wightman Rindler-wedge KMS state.

The choice of H is the load-bearing structural input. We make the canonical choice, motivated by the alignment with the Wightman BW vacuum (Section 6 below):

$$H_{\text{local}} \;:=\; K_\alpha^W \,/\, \beta, \tag{11}$$

where K_α^W is the geometric BW-α generator of Definition 5.1 below. With this choice, $\beta H_{\text{local}} = K_\alpha^W$ is itself β-independent at the algebra-action level, so the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ is the same for every β. This is a deliberate construction-level choice; an alternative choice $H_{\text{local}} = D_W$ (the wedge-restricted Camporesi–Higuchi Dirac) would lift the $\beta = 2\pi$ period to the spinor double-cover $-I$ rather than the identity operator $+I$ of Section 5 (the conjugation flow still closes, but at the double-cover sign), isolating (11) as a derived structural finding. We return to this point in Section 7.

4.3 Four-witness unification and unit normalisation

The four physical witnesses (Hartle–Hawking, Sewell, Bisognano–Wichmann, Unruh) share the structural identity

$$\beta_W \;=\; 2\pi \,/\, \kappa_g(W), \tag{12}$$

where $\kappa_g(W)$ is the surface gravity of the bifurcate Killing horizon bounding the causal region W: $\kappa_g^{\text{HH}} = 1/(4M)$ for the Schwarzschild exterior of mass M, giving $\beta^{\text{HH}} = 8\pi M$; $\kappa_g^{\text{Sew}} = \kappa_g^{\text{HH}}$ for the abstract KMS-modular reading of the Hartle–Hawking state; $\kappa_g^{\text{BW}} = 1$ in natural rapidity units for the Rindler wedge, giving $\beta^{\text{BW}} = 2\pi$; $\kappa_g^{\text{Unruh}} = a$ for the Rindler wedge in proper-acceleration units, giving $\beta^{\text{Unruh}} = 2\pi/a$.

Unit normalisation. We adopt the BW rapidity-units convention $\kappa_g = 1$ (denoted *U-1* in the architectural scoping) as the canonical witness. The four physical witnesses are then instantiated as parameterised BW with witness-specific κ_g values: BW $\to \kappa_g = 1$, HH $\to \kappa_g = 1/(4M)$, Sew $\to \kappa_g = 1/(4M)$, Unruh $\to \kappa_g = a$. The crucial structural observation (Sections 5–8) is that the period closure $\sigma_{2\pi}(O) = O$ at the operator level is *independent of* κ_g, because the geometric BW-α generator has integer spectrum on the spinor basis and the closure $e^{i\cdot 2\pi\cdot n} = 1$ for integer n does not depend on the overall rescaling. The four-witness theorem therefore collapses to a single operator-system test at the U-1 normalisation; the parameterisations to HH, Sewell, and Unruh follow by a rescaling that is invisible at the period.

5 The geometric construction (BW-α)

The first construction of the modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ is the geometric BW-α realization, obtained by identifying the wedge-preserving Killing vector on the Riemannian hemisphere with the spatial rotation generator around the Hopf-base axis.

5.1 The geometric generator

Definition 5.1 (BW-α geometric generator on the wedge)**.** The *BW-α geometric generator* on the wedge sub-Hilbert space $\mathcal{H}_W = P_W \mathcal{H}_{n_{\max}}$ is the operator

$$K_\alpha^W \;:=\; \operatorname{diag}\bigl(\text{two\_}m_j\bigr) \quad \text{on} \quad \mathcal{H}_W, \tag{13}$$

acting on each wedge basis state $\psi_{n,l,|m_j|,\chi}$ (with m_j chosen positive on the wedge P_W) by multiplication by two_m_j, the doubled m-quantum-number.

Remark 5.2 (Why two_m_j rather than m_j)**.** On the full-Dirac spinor basis $\{\psi_{n,l,m_j,\chi}\}$, the quantum number m_j takes half-integer values $\{-l-1/2, -l+1/2, \ldots, l+1/2\}$. To obtain an integer spectrum — the load-bearing property for the bit-exact period closure of Theorem 5.4 — we work with the doubled generator two_$m_j = 2m_j$, taking odd integer values. This is the standard half-integer-spin doubling convention from representation theory of SU(2). The factor of 2 does not affect the modular flow at the period (modular flow is invariant under overall rescaling of K by integer factors), but it makes the integer-spectrum property manifest.

Proposition 5.3 (Integer spectrum)**.** *The spectrum of K_α^W on $\mathcal{H}_W$ is contained in the odd integers $\{-(2l+1), -(2l-1), \ldots, +(2l-1), +(2l+1)\}$ over $l \in \{0, 1, \ldots, n_{\max}-1\}$. In particular,* $\mathrm{Spec}(K_\alpha^W) \subset \mathbb{Z}$.

Proof. On the wedge P_W, m_j takes values $\{1/2, 3/2, \ldots, l+1/2\}$ (positive half-integers up to $l+1/2$). The doubled generator two_m_j takes values $\{1, 3, \ldots, 2l+1\}$, all odd positive integers. Including the wedge-anti-wedge structure via the full-Dirac basis would give the full range $\{-(2l+1), \ldots, +(2l+1)\}$ in odd integer steps; on the wedge itself we have the positive sub-range. $\square$

5.2 The geometric modular flow

The geometric BW-α modular flow on the wedge sub-operator-system $\mathcal{O}_{W,n_{\max}}$ is the inner-derivation flow

$$\sigma_t^\alpha(O) \;:=\; e^{itK_\alpha^W}\, O\, e^{-itK_\alpha^W}, \qquad O \in \mathcal{O}_{W,n_{\max}},\; t \in \mathbb{R}. \tag{14}$$

This is well-defined as a strongly continuous one-parameter group of unital $*$-automorphisms on $B(\mathcal{H}_W)$. Its restriction to $\mathcal{O}_{W,n_{\max}}$ may leave the operator system (since $\mathcal{O}_{W,n_{\max}}$ is not multiplicatively closed); we return to this point in Section 5.3.

Theorem 5.4 (BW-α period closure at finite cutoff)**.** *For every $n_{\max} \geq 1$ and every $O \in B(\mathcal{H}_W)$, the BW-α modular flow* (14) *satisfies the period closure*

$$\sigma_{2\pi}^\alpha(O) \;=\; O \tag{15}$$

identically, i.e., as a bit-exact operator identity.

Proof. By Proposition 5.3, K_α^W is diagonal with integer eigenvalues $\{n_1, n_2, \ldots, n_{\dim_W}\}$ in its eigenbasis. The unitary $e^{i\cdot 2\pi\cdot K_\alpha^W}$ is then diagonal with eigenvalues $e^{i\cdot 2\pi\cdot n_j} = 1$ for every j (since each n_j is an integer): $e^{i\cdot 2\pi\cdot K_\alpha^W} = I_{\dim_W}$ identically. Hence $\sigma_{2\pi}^\alpha(O) = I\cdot O\cdot I = O$ for every O. $\square$

Numerical verification. Theorem 5.4 is an exact algebraic identity at the level of the symbolic construction. In a floating-point computation, the matrix exponential $e^{i\cdot 2\pi\cdot K_\alpha^W}$ is computed by diagonalisation and then exponentiation of the eigenvalues, so the identity holds to within the floating-point precision of the diagonalisation step. We have computed the maximum periodicity residual

$$r_W^\alpha(O) \;:=\; \|\sigma_{2\pi}^\alpha(O) - O\|_F \tag{16}$$

for the first five non-identity multipliers $O \in \mathcal{O}_{n_{\max}}$ (restricted to the wedge), at $n_{\max} \in \{2, 3, 4, 5\}$. The results are recorded in Table 1.

The $n_{\max} = 3$ row is structurally distinguished: $\dim \mathcal{H}_3 = 40 = g_3^{\mathrm{Dirac}} = \Delta^{-1}$, the structurally distinguished selection-principle cutoff of the present author's Paper 2 [32] fine-structure-constant observation. The L1 closure at this cutoff is the operator-system-level confirmation that the M1 mechanism's 2π closure holds on the Paper-2-relevant truncation.

Table 1: Maximum BW-α periodicity residual $\max_O r_W^\alpha(O)$ at finite cutoff. Computed in double-precision floating point on the full-Dirac basis; residuals scale as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\text{machine}})$, i.e., pure round-off accumulation with no structural drift. Source data: `debug/data/l1_modular_hamiltonian_results.json`.

$n_{\max}$	$\dim \mathcal{H}_{n_{\max}}$	$\dim_W$	$\max_O r_W^\alpha(O)$	verdict
2	16	8	1.80×10^{-16}	STRONG_IDENTIFICATION
3	40	20	4.76×10^{-16}	STRONG_IDENTIFICATION
4	80	40	9.33×10^{-16}	STRONG_IDENTIFICATION
5	140	70	1.59×10^{-15}	STRONG_IDENTIFICATION

5.3 Operator-system non-leakage

A separate concern flagged in the architectural scoping (`debug/l1_modular_hamiltonian_architecture_memo.` §10a) is that the truncated operator system $\mathcal{O}_{n_{\max}}$ is $*$-closed but not multiplicatively closed: the modular flow σ_t^α acts on all of $B(\mathcal{H}_W)$, and its restriction to $\mathcal{O}_{W,n_{\max}}$ might leak out. The integer-spectrum closure Theorem 5.4 resolves this without further work: $\sigma_{2\pi}^\alpha(O) = O$ for every O, so the flow at the period $t = 2\pi$ leaves every element of $\mathcal{O}_{W,n_{\max}}$ inside $\mathcal{O}_{W,n_{\max}}$ trivially. At intermediate $t \in (0, 2\pi)$ the flow may leave the operator system, but the period closure (the modular identity we test) is unaffected.

6 The Tomita–Takesaki construction (BW-γ)

The second construction of the modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ is the Tomita–Takesaki BW-γ realization, obtained by polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space of the wedge-restricted Gibbs state.

6.1 The GNS Hilbert–Schmidt space

Let $\rho := e^{-K_\alpha^W}/Z$ be the wedge density matrix (Section 4.2 with $H_{\text{local}} = K_\alpha^W/\beta$, so $\beta H_{\text{local}} = K_\alpha^W$ is β-independent). For the von Neumann algebra $\mathcal{A}_W := B(\mathcal{H}_W) \cong M_{\dim_W}(\mathbb{C})$ (Type $\mathrm{I}_{\dim_W}$ factor) and the faithful normal state $\omega(O) := \mathrm{Tr}_{\mathcal{H}_W}(\rho\, O)$, the GNS triple $(\pi_\omega, \mathcal{H}_{\text{GNS}}, \Omega)$ is realised on the Hilbert–Schmidt space

$$\mathcal{H}_{\text{GNS}} \;=\; M_{\dim_W}(\mathbb{C}) \;\cong\; \mathbb{C}^{\dim_W^2}, \qquad \langle X, Y\rangle_{\text{GNS}} \;:=\; \mathrm{Tr}(X^* Y), \tag{17}$$

with cyclic vector $\Omega := \rho^{1/2}$ (the canonical purification of ρ). The left action of $a \in \mathcal{A}_W$ is $L_a(X) = aX$, the right action of a from the commutant $\mathcal{A}_W^{\text{op}}$ is $R_a(X) = Xa$. Numerically, $\dim \mathcal{H}_{\text{GNS}} = \dim_W^2 = 64, 400, 1600, 4900$ at $n_{\max} = 2, 3, 4, 5$.

6.2 The modular S operator and polar decomposition

The Tomita modular operator $S : \pi_\omega(a)\Omega \mapsto \pi_\omega(a^*)\Omega$ is the antilinear closure of the conjugate-linear involution $a\Omega \mapsto a^*\Omega$ on the cyclic-separating orbit. In the Hilbert–Schmidt realisation (17), the cyclic orbit is the set of $X = a\rho^{1/2}$ for $a \in \mathcal{A}_W$ (equivalently, all of $\mathcal{H}_{\text{GNS}}$ if ρ is positive definite, which is the present case). The operation $a\Omega \mapsto a^*\Omega$ is then

$$S(X) \;=\; \rho^{1/2}\,(X\rho^{-1/2})^* \;=\; \rho^{1/2}\,\rho^{-1/2}\,X^* \;=\; X^*, \tag{18}$$

where the second equality uses $\rho^{1/2}\rho^{-1/2} = I$ and the fact that ρ is self-adjoint, so the $\rho^{1/2}$ on the left and $\rho^{-1/2}$ on the right cancel against X^* on a tracial cyclic vector. The polar decomposition $S = J_{\text{TT}} \cdot \Delta^{1/2}$ gives:

$$\Delta \;=\; S^* S, \qquad \Delta(X) \;=\; \rho\, X\, \rho^{-1} \tag{19}$$

acting on $X \in \mathcal{H}_{\rm GNS}$, and

$$J_{\rm TT}(X) \;=\; \rho^{1/2}\, X^*\, \rho^{-1/2} \tag{20}$$

acting antilinearly on $\mathcal{H}_{\rm GNS}$. In the column-stacked matrix form $\mathrm{vec}(X) \in \mathbb{C}^{\dim_W^2}$, the modular operator Δ is

$$\Delta \;=\; (\rho^{-1})^T \otimes \rho \quad \text{acting on} \quad \mathrm{vec}(X) \in \mathbb{C}^{\dim_W^2}. \tag{21}$$

The *modular Hamiltonian* is then

$$K_{\rm TT} \;:=\; -\log \Delta, \tag{22}$$

and the *modular flow on the algebra* is

$$\sigma_t^{\rm TT}(a) \;:=\; \Delta^{it}(a) \;=\; \rho^{it}\, a\, \rho^{-it} \;=\; e^{-it\, K_\alpha^W}\, a\, e^{+it\, K_\alpha^W}, \tag{23}$$

where the last equality uses $\rho = e^{-K_\alpha^W}/Z$ and the Z factor cancels in the conjugation.

6.3 The Tomita antilinear conjugation

A structurally important caution is that the Tomita modular conjugation $J_{\rm TT}$ defined in (20) is *categorically different* from the Connes KO-dim 3 charge conjugation $J_{\rm GV}$ on the underlying Camporesi–Higuchi spectral triple.

Proposition 6.1 ($J_{\rm TT} \neq J_{\rm GV}$)**.** *The Tomita conjugation $J_{\rm TT}$ on $\mathcal{H}_{\rm GNS}$ and the Connes real structure $J_{\rm GV}$ on $\mathcal{H}_{n_{\max}}$ are categorically distinct antilinear operators with the following signature distinction:*

$$J_{\rm TT}^2 \;=\; +I, \qquad J_{\rm GV}^2 \;=\; -I. \tag{24}$$

Proof. For $J_{\rm TT}^2 = +I$: applying (20) twice gives

$$J_{\rm TT}^2(X) \;=\; J_{\rm TT}\big(\rho^{1/2}\, X^*\, \rho^{-1/2}\big) \;=\; \rho^{1/2}\, (\rho^{1/2}\, X^*\, \rho^{-1/2})^*\, \rho^{-1/2} \;=\; \rho^{1/2}\, \rho^{-1/2}\, X\, \rho^{1/2}\, \rho^{-1/2} \;=\; X,$$

where the second-to-last step uses $(\rho^{1/2})^* = \rho^{1/2}$ and $(\rho^{-1/2})^* = \rho^{-1/2}$ (both are self-adjoint), and the last step uses $\rho^{1/2}\rho^{-1/2} = I$ twice. For $J_{\rm GV}^2 = -I$: the Connes real structure on a KO-dim 3 spectral triple satisfies $J^2 = -I$ by convention (cf. Paper 32 § IV `rem:axiom_audit`). □

The two antilinear conjugations coexist on the same Hilbert spaces: $J_{\rm GV}$ on $\mathcal{H}_{n_{\max}}$, $J_{\rm TT}$ on $\mathcal{H}_{\rm GNS} \cong B(\mathcal{H}_W)$. Conflating them (e.g., by importing `geovac.real_structure.RealStructure` into the modular-Hamiltonian construction) would silently produce algebraic errors, since $J_{\rm TT}^2 - (-I) = 2I \neq 0$.

Remark 6.2 (State-dependence vs. kinematic intrinsicness)**.** $J_{\rm GV}$ is intrinsic to the spectral triple at construction time: once $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ is specified, $J_{\rm GV}$ is fixed by the KO-dim convention. $J_{\rm TT}$ is *state-dependent*: each KMS state ω defines its own $J_{\rm TT}^{(\omega)}$ via Tomita's theorem on the GNS triple of ω. The state-dependence is consistent with the standard Tomita–Takesaki theorem [26, 27].

6.4 Bit-exact period closure at the Tomita level

Theorem 6.3 (BW-γ period closure at finite cutoff)**.** *For every $n_{\max} \geq 1$ and every $a \in B(\mathcal{H}_W)$, the Tomita–Takesaki modular flow (23) satisfies the period closure*

$$\sigma_{2\pi}^{\rm TT}(a) \;=\; a \tag{25}$$

identically, with maximum periodicity residual

$$r_W^{\rm TT}(O) \;:=\; \|\sigma_{2\pi}^{\rm TT}(O) - O\|_F$$

scaling as $O(\sqrt{\dim_W} \cdot \varepsilon_{\rm machine})$ in floating-point computation.

Proof. From (23), $\sigma_{2\pi}^{\rm TT}(a) = e^{-i\cdot 2\pi\cdot K_\alpha^W}\, a\, e^{+i\cdot 2\pi \cdot K_\alpha^W}$. By Proposition 5.3, K_α^W has integer spectrum, so $e^{\pm i\cdot 2\pi\cdot K_\alpha^W} = I$ identically. Hence $\sigma_{2\pi}^{\rm TT}(a) = a$. □

Numerical verification. In floating-point computation, the modular operator Δ is built on the GNS Hilbert–Schmidt space $\mathcal{H}_{\mathrm{GNS}} \cong \mathbb{C}^{\dim_W^2}$ via the tensor-product form (21), with each $\rho^{\pm 1}$ assembled by matrix exponential of $\pm K_\alpha^W$ and a numerical division by Z. The modular flow σ_t^{TT} is then computed by acting Δ^{it} on each input operator a written in vec form, then reshaping. We have computed the maximum periodicity residual $r_W^{\mathrm{TT}}(O)$ at $n_{\max} \in \{2, 3, 4, 5\}$; results are in Table 2.

Table 2: Maximum BW-γ Tomita-level periodicity residual $\max_O r_W^{\mathrm{TT}}(O)$ at finite cutoff, computed via polar decomposition on the GNS Hilbert–Schmidt space. The slightly larger residual at $n_{\max} = 5$ (vs. Table 1) is pure numerical conditioning from the $\dim_W^2 \times \dim_W^2 = 4900 \times 4900$ matrix diagonalisation of Δ with extreme eigenvalue ratios λ_i/λ_j; no structural drift. Source data: `debug/data/l1_tighten_tomita_results.json`.

$n_{\max}$	$\dim_W$	$\dim \mathcal{H}_{\mathrm{GNS}}$	$\max_O r_W^{\mathrm{TT}}(O)$	verdict
2	8	64	1.09×10^{-16}	UNIFIED_STRONG
3	20	400	1.07×10^{-15}	UNIFIED_STRONG
4	40	1600	3.03×10^{-15}	UNIFIED_STRONG
5	70	4900	4.79×10^{-15}	UNIFIED_STRONG

The $J_{\mathrm{TT}}^2 = +I$ identity of Proposition 6.1 is verified numerically at every $n_{\max}$ to machine precision (residuals $\leq 7 \times 10^{-17}$): this distinguishes J_{TT} from J_{GV} at the genuine state-dependent level, not just at the canonical $U = I$ sanity case.

6.5 Spectrum of K_{TT}

Proposition 6.4 (Spectrum of K_{TT} on $\mathcal{H}_{\mathrm{GNS}}$)**.** *The eigenvalues of $K_{\mathrm{TT}} = -\log \Delta$ on $\mathcal{H}_{\mathrm{GNS}}$ are $\{n_j - n_k : 1 \leq j, k \leq \dim_W\}$, where $\{n_j\}$ is the integer spectrum of K_α^W. In particular:*

(a) $\mathrm{Spec}(K_{\mathrm{TT}}) \subset \mathbb{Z}$ *(integer spectrum on $\mathcal{H}_{\mathrm{GNS}}$);*

(b) K_{TT} *has at least* $\dim_W$ *zero eigenvalues (from the diagonal $j = k$ pairs);*

(c) The maximum eigenvalue of K_{TT} is $2\max_j n_j = 2(2(n_{\max} - 1) + 1) = 2(2n_{\max} - 1)$; for $n_{\max} = 2$ this is ≤ 4, and the maximum is exactly 2 on the wedge (since $\max_W \mathrm{two\_}m_j = +1$ at $n_{\max} = 2$).

Proof. By (21), $\Delta = (\rho^{-1})^T \otimes \rho$ in the vec representation, with $\rho = e^{-K_\alpha^W}/Z$. The eigenvalues of ρ are $\{e^{-n_j}/Z\}_{j=1}^{\dim_W}$, so the eigenvalues of Δ are $\{e^{n_k} \cdot e^{-n_j}\}_{j,k} = \{e^{n_k - n_j}\}_{j,k}$. Hence $K_{\mathrm{TT}} = -\log \Delta$ has eigenvalues $\{n_j - n_k\}$, which are integer differences of the integer spectrum of K_α^W. The diagonal $j = k$ pairs contribute $n_j - n_j = 0$, giving at least $\dim_W$ zero eigenvalues (more if K_α^W has spectral degeneracies). $\square$

The integer-spectrum property of Proposition 6.4(a) is the structural reason the bit-exact closure of Theorem 6.3 holds: $e^{i \cdot 2\pi \cdot K_{\mathrm{TT}}} = I$ on $\mathcal{H}_{\mathrm{GNS}}$ by $e^{i \cdot 2\pi \cdot m} = 1$ for integer m, so the modular flow Δ^{it} acting at $t = 2\pi$ collapses to the identity, and $\sigma_{2\pi}^{\mathrm{TT}}(a) = I \cdot a \cdot I = a$ for every a. This is the same mechanism as Theorem 5.4 but realised at the GNS-Hilbert–Schmidt level.

6.6 Propinquity rate cross-check

The closure of Theorem 6.3 is a finite-cutoff identity, not a propinquity limit statement. To confirm consistency with Paper 38's qualitative-rate state-space-GH bound (7), we compute the ratio

$$\frac{\max_O r_W^{\mathrm{TT}}(O)}{\gamma_{n_{\max}}^{(\mathrm{L2})}}$$

where $\gamma^{(\mathrm{L2})}_{n_{\max}}$ is the L2 mass-concentration moment of the central spectral Fejér kernel on SU(2) ([37, Lemma L2]): $\gamma_2^{(\mathrm{L2})} = 2.0746$, $\gamma_3^{(\mathrm{L2})} = 1.6101$, $\gamma_4^{(\mathrm{L2})} = 1.3223$, $\gamma_5^{(\mathrm{L2})} = 1.1302$. The ratios are 8.7×10^{-17}, 3.0×10^{-16}, 7.1×10^{-16}, 1.4×10^{-15}: several orders of magnitude below the qualitative-rate bound at every tested $n_{\max}$. The L1 closure is therefore *much stronger* than the qualitative-rate propinquity bound would predict at finite cutoff; the closure does not require the GH limit and is structural rather than asymptotic.

7 Unified closure

The two constructions of Sections 5 and 6 are not the same operator on $\mathcal{H}_W$ versus $\mathcal{H}_{\mathrm{GNS}}$ (the GNS construction lives on a larger Hilbert space). But the modular flows they generate on the algebra $B(\mathcal{H}_W)$ are intimately related, and at the period $t = 2\pi$ they collapse to the identical operator-system identity. This section makes the relation precise.

7.1 Flow conjugacy

Theorem 7.1 (BW-α / BW-γ flow conjugacy)**.** *For every $a \in B(\mathcal{H}_W)$ and every $t \in \mathbb{R}$,*

$$\sigma_t^{\mathrm{TT}}(a) \;=\; \sigma_{-t}^{\alpha}(a), \tag{26}$$

where σ_t^{α} is the geometric BW-α flow (14) *and σ_t^{TT} is the Tomita–Takesaki BW-γ flow* (23)*.*

Proof. By (23), $\sigma_t^{\mathrm{TT}}(a) = e^{-itK_\alpha^W}\, a\, e^{+itK_\alpha^W}$. By (14), $\sigma_{-t}^{\alpha}(a) = e^{-itK_\alpha^W}\, a\, e^{+itK_\alpha^W}$. The two expressions are literally identical. □

Consequence at the period. At $t = 2\pi$, equation (26) reads $\sigma_{2\pi}^{\mathrm{TT}}(a) = \sigma_{-2\pi}^{\alpha}(a) = \sigma_{2\pi}^{\alpha}(a)^{-1}$, where the last equality uses that the flow σ_t^{α} is a one-parameter group with the inverse at $t = 2\pi$ being the flow at $t = -2\pi$. But by Theorem 5.4, $\sigma_{2\pi}^{\alpha}(a) = a = a^{-1}$ trivially (this is an operator equation, not a group inverse), so $\sigma_{2\pi}^{\mathrm{TT}}(a) = a$ as well, consistent with Theorem 6.3. The two closures are not just both true; they are the *same* closure under the conjugacy (26).

Numerical confirmation at general t. At a general non-period t (e.g., $t = 1$), the two flows σ_t^{TT} and σ_t^{α} are distinct (the conjugacy (26) relates σ_t^{TT} to σ_{-t}^{α}, not to σ_t^{α}). We have verified this numerically: the average cross-flow difference $\langle \|\sigma_1^{\mathrm{TT}}(O) - \sigma_1^{\alpha}(O)\|_F \rangle_O$ over the same five test operators is ≈ 0.16 at $n_{\max} = 2$, clearly non-zero. The sign-flipped comparison $\langle \|\sigma_1^{\mathrm{TT}}(O) - \sigma_{-1}^{\alpha}(O)\|_F \rangle_O$ gives $\approx 4 \times 10^{-17}$, machine precision. The conjugacy (26) is therefore verified bit-exactly at general t, not just at the period.

7.2 The derived-Hamiltonian finding

The unified closure rests on the deliberate choice $H_{\mathrm{local}} = K_\alpha^W/\beta$ (equation (11)) for the local Hamiltonian generating the wedge KMS state. This is a construction-level choice, not a derivation: we wrote down the wedge KMS state as $\rho_W = e^{-K_\alpha^W}/Z$ rather than (say) $\rho_W = e^{-\beta D_W}/Z$ for the wedge-restricted Camporesi–Higuchi Dirac. The motivation is structural (we want the modular Hamiltonian's period to match the canonical BW $\beta = 2\pi$), but it is *a* choice of local Hamiltonian, not the unique choice.

Two consequences are immediate, and together they isolate the choice as a derived structural finding rather than a uniqueness theorem:

(I) Matches the Wightman BW vacuum. In the continuum Bisognano–Wichmann theorem, the local Hamiltonian on the Rindler wedge is the boost generator K_{boost}, and the Wightman vacuum restricted to the wedge algebra is the Gibbs state $\rho_W = e^{-2\pi K_{\text{boost}}}/Z$ at inverse temperature $\beta = 2\pi$. Our choice $H_{\text{local}} = K_\alpha^W/\beta$ matches this structure exactly: $\beta H_{\text{local}} = K_\alpha^W$ is the wedge-restricted boost-class generator (the spatial rotation generator around the Hopf-base axis, on the Riemannian side), and the KMS state is $\rho_W = e^{-K_\alpha^W}/Z = e^{-2\pi K_\alpha^W/(2\pi)}/Z$, the operator-system analog of the Wightman BW vacuum. The choice is therefore the canonical operator-system analog of the Wightman state, not a post-hoc engineering choice.

(II) The framework's intrinsic Dirac is not the right local Hamiltonian. If we had chosen $H_{\text{local}} = D_W$ (the wedge-restricted Camporesi–Higuchi Dirac), then $\beta H_{\text{local}} = 2\pi D_W$ would have half-integer spectrum on the spinor basis (the eigenvalues of D_W are $\chi \cdot (n + 1/2)$, with $n + 1/2$ half-integer), so the *operator-level* period lift $e^{i\cdot 2\pi\cdot\beta D_W} = -I$ at $\beta = 2\pi$ — the global -1 of the spinor double cover — whereas the boost-class K_α^W has odd-integer spectrum and $e^{i\cdot 2\pi\cdot K_\alpha^W} = +I$, the identity operator. We emphasise the precise sense of the distinction: the modular flow $\sigma_t(\cdot) = U_t(\cdot)U_t^{-1}$ acts by *conjugation*, in which a global scalar phase cancels, so the flow-level period closure $\sigma_{2\pi}(O) = O$ still holds with D_W (the $-I$ cancels: $(-I)\,O\,(-I) = O$; we verified this bit-exactly, residual $< 10^{-13}$ at $n_{\max} = 2, 3, 4$). The distinction is therefore *not* at the flow level but at the operator level: only K_α^W lifts the $\beta = 2\pi$ period to the identity operator $+I$; D_W lifts it to the double-cover $-I$.

This is a non-trivial structural finding. The framework's intrinsic Dirac D_{CH} is the canonical spectral-action object on the Camporesi–Higuchi triple, and one might naively expect it to play the role of the local Hamiltonian for any natural KMS state on the truncated operator system. The above shows that, while either choice generates a 2π-periodic conjugation flow, only the boost-class generator K_α^W realises the canonical BW vacuum with the identity-operator period lift at $\beta = 2\pi$; the wedge-restricted Dirac D_W carries the spinor double-cover sign instead. The two play complementary roles: D_{CH} is the spectral-action object (the Connes–Chamseddine input for the heat-kernel expansion); K_α^W is the wedge-modular-Hamiltonian object (the Tomita–Takesaki input for the modular-flow construction). On the round S^3 they happen to agree up to an overall scale (both are diagonal in the spinor basis with simple spectral structure), but only one of them has the integer spectrum that lifts the $\beta = 2\pi$ period to the identity operator (the same integer-spectrum / $e^{2\pi i K} = +I$ structure that underlies the compact-boost reading of Section 10).

Remark 7.2 (A speculative reading)**.** We note, but do not pursue here, the possibility that the distinction between the spectral-action Dirac D_{CH} and the modular-Hamiltonian generator K_α^W is structurally significant for the framework's interpretation. In the spectral-action paradigm (Chamseddine–Connes [7]), the Dirac is the universal object from which all physics is read; in the thermal-time paradigm (Connes–Rovelli [9]), the modular Hamiltonian of an algebraic state is the physical-time generator. On the round S^3 Camporesi–Higuchi triple at the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$, these are not the same operator. The structural status of this distinction — whether it is a peculiarity of the round S^3 truncation or a deeper feature of the framework's spectral content — is open. We flag it as one of the open questions in Section 10.

7.3 Cross-witness collapse at the algebra-action level

The four-witness collapse mentioned in Section 4.3 acquires a sharper form at the algebra-action level. With $H_{\text{local}} = K_\alpha^W/\beta$, the density matrix is

$$\rho_W = \frac{e^{-\beta H_{\text{local}}}}{Z} = \frac{e^{-K_\alpha^W}}{Z} \tag{27}$$

independent of β. The Tomita modular operator Δ constructed from ρ_W via (21) and the modular Hamiltonian $K_{\mathrm{TT}} = -\log \Delta$ are therefore also β-independent. The four-witness theorem at the operator-system level reduces to the assertion that all four physical witnesses (BW, HH, Sew, Unruh) produce the *same* modular operator Δ and the same modular Hamiltonian K_{TT}, with the witness-specific κ_g values dropping out of the algebra-action construction.

We have verified this numerically at $n_{\max} \in \{2, 3, 4, 5\}$ across the six witness instantiations (BW canonical at $\kappa_g = 1$; Hartle–Hawking at $M = 1$, $\kappa_g = 1/4$; Hartle–Hawking at $M = 2$, $\kappa_g = 1/8$; Sewell at $M = 1$, $\kappa_g = 1/4$; Unruh at $a = 1$, $\kappa_g = 1$; Unruh at $a = 2$, $\kappa_g = 2$). The cross-witness consistency residual is exactly 0 at every tested $n_{\max}$ between any pair of witnesses, confirming the β-independence of the density matrix at the construction level.

8 Four-witness collapse and a single construction

The numerical content of the four-witness collapse is summarised in Table 3 and the structural content is the following corollary.

Table 3: Cross-witness consistency at the BW-α level (left columns) and at the BW-γ Tomita level (right columns). Cross-consistency residual is the difference between the period residual of each witness and the BW canonical period residual at the same $n_{\max}$: exactly 0 across all six witnesses at every tested $n_{\max}$. Source data: `debug/data/l1_modular_hamiltonian_results.json`, `debug/data/l1_tighten_tomita_results.json`.

$n_{\max}$	BW ($\kappa_g=1$)	$\mathrm{HH}_{M=1}$ ($\kappa_g=1/4$)	$\mathrm{HH}_{M=2}$ ($\kappa_g=1/8$)	$\mathrm{Sew}_{M=1}$ ($\kappa_g=1/4$)	$\mathrm{Unruh}_{a=1}$ ($\kappa_g=1$)	$\mathrm{Unruh}_{a=2}$ ($\kappa_g=2$)
BW-α residual (Table 1):						
2	1.80e-16	1.80e-16	1.80e-16	1.80e-16	1.80e-16	1.80e-16
3	4.76e-16	4.76e-16	4.76e-16	4.76e-16	4.76e-16	4.76e-16
4	9.33e-16	9.33e-16	9.33e-16	9.33e-16	9.33e-16	9.33e-16
5	1.59e-15	1.59e-15	1.59e-15	1.59e-15	1.59e-15	1.59e-15
BW-γ Tomita residual (Table 2):						
2	1.09e-16	1.09e-16	1.09e-16	1.09e-16	1.09e-16	1.09e-16
3	1.07e-15	1.07e-15	1.07e-15	1.07e-15	1.07e-15	1.07e-15
4	3.03e-15	3.03e-15	3.03e-15	3.03e-15	3.03e-15	3.03e-15
5	4.79e-15	4.79e-15	4.79e-15	4.79e-15	4.79e-15	4.79e-15

Corollary 8.1 (Single operator-system construction for the four witnesses)**.** *Under the unit normalisation U-1 and the wedge KMS state choice* $\rho_W = e^{-K_\alpha^W}/Z$*, the operator-system-level realisations of the Hartle–Hawking, Sewell, Bisognano–Wichmann, and Unruh witnesses on the truncated Camporesi–Higuchi triple* $\mathcal{T}_{n_{\max}}$ *collapse to a single construction. The modular operator* Δ *and the modular Hamiltonian* K_{TT} *are bit-identical across the six witness instantiations. The witness-specific physics enters only through the parameterisation of* κ_g*, which is invisible at the algebra-action level and at the period closure* $\sigma_{2\pi}(O) = O$.

Proof. The density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is β-independent (Section 7.3). The Tomita modular operator Δ on the GNS Hilbert–Schmidt space is constructed entirely from ρ_W via (21), and the modular Hamiltonian $K_{\mathrm{TT}} = -\log \Delta$ depends only on Δ. Both are therefore β-independent. β-independence at the algebra-action level immediately implies identical period closure at $t = 2\pi$ for all six witness instantiations. $\square$

Structural reading. At the operator-system level on the truncated Camporesi–Higuchi triple, the four-witness Wick-rotation theorem reduces to one construction. The witness-specific physical content (mass of the black hole, proper acceleration of the Rindler observer) parameterises the correspondence to the continuum physical observable but does not modify the underlying modular-Hamiltonian construction. The same 2π that appears as the Hopf-base measure factor $\mathrm{Vol}(S^2)/\pi^2$ in the Paper 38 state-space-GH rate constant (7) also drives the modular period closure of Theorems 5.4–6.3.

This is the operator-system-level realisation of the M1 sub-mechanism of the master Mellin engine (Paper 32 § VIII case-exhaustion theorem of the present author's framework). The metric-functional-level reading ([35, Rem. rem:bisognano_wichmann_reading], Sprint TD Track 4, Sprint Unruh-pendant) identifies the framework's 2π on a temporal S^1_τ with the modular-flow / boost-orbit period of the four witnesses via the published Wick-rotation chain. The present paper lifts that identification to the operator-system level inside the spectral triple, eliminating the dependence on the Wick-rotation prescription for the period-closure identity itself: the 2π in the modular period is now a framework-internal output of the spectral truncation, not an externally imposed Wick-rotation matching.

Remark 8.2 (Pythagorean orthogonality underlies the six-witness collapse)**.** A post-construction PSLQ probe on the Lorentzian extension ([40, § 10.2, Cor. cor:pythagorean_orthogonality]; `debug/h_local_residual_pslq_memo.md`, `debug/six_witness_hs_orthogonality_memo.md`) identified a structural inner-product identity that underlies the six-witness collapse of Corollary 8.1. For every witness $w \in \{\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2}\}$, the BW-aligned local Hamiltonian $H^{(w)}_{\mathrm{local}} := K^W_\alpha/\beta^{(w)}$ and the wedge-restricted Camporesi–Higuchi Dirac D^{GV}_W are Hilbert–Schmidt orthogonal:

$$\langle H^{(w)}_{\mathrm{local}}, D^{\mathrm{GV}}_W \rangle_{\mathrm{HS}} \;:=\; \mathrm{Tr}\big((H^{(w)}_{\mathrm{local}})^\dagger\, D^{\mathrm{GV}}_W\big) \;=\; 0,$$

verified bit-exactly at every cell of the 18-cell panel (six witnesses $\times$ $n_{\max} \in \{2,3\}$; extended to $n_{\max} \in \{1,\dots,5\}$ on the BW canonical row; $\max|\langle H, D\rangle_{\mathrm{HS}}| = 8.9 \times 10^{-16}$, machine precision). The witness-independence of the inner product is a κ_g-linearity corollary: $H^{(w)}_{\mathrm{local}}$ scales linearly with $\kappa_g^{(w)}$, while D^{GV}_W is witness-independent.

The mechanism is structural rather than computational: in the unfolded wedge spinor basis, $H^{(w)}_{\mathrm{local}}$ is real and diagonal (eigenvalues $\mathrm{two\_m}_j > 0$, the rotation generator around the Hopf axis), while D^{GV}_W is off-diagonal (it intertwines $\pm m_j$ chirality partners via the wedge basis change). The two operators occupy mutually-orthogonal subspaces of $\mathcal{B}(\mathcal{H}_W)$, so the inner product vanishes; the diagonal/off-diagonal subspace decomposition is sketched at the basis level and supports a Pythagorean form for the residual

$$\big\|H^{(w)}_{\mathrm{local}} - D^{\mathrm{GV}}_W\big\|_F^2 \;=\; \frac{(\kappa_g^{(w)})^2 \cdot S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

with S, D pure rationals in $n_{\max}$ (Paper 43 § 10.2). A formal proof of the subspace decomposition at general $n_{\max}$ on the truncated spectral triple is a named follow-on; the empirical verification across the 18-cell panel is strong evidence the orthogonality is structural rather than coincidental.

Reading. Corollary 8.1 states that the six witnesses share a single modular operator and a single modular Hamiltonian via β-independence of ρ_W. The present remark sharpens that reading at the inner-product level: the period closure $\sigma_{2\pi}(O) = O$ for every O in the operator system is underlaid by a structural Pythagorean identity that controls how $H^{(w)}_{\mathrm{local}}$ and D^{GV}_W decompose orthogonally in HS norm, with the witness-specific physical content carried by the κ_g-dependence of the M1 piece and not by any rotation between the two subspaces.

9 Related work and scope

Connes–Rovelli thermal time hypothesis. The closest published analog of the present construction is the Connes–Rovelli thermal time hypothesis [9], which identifies the Tomita modular flow of a general-covariant von Neumann algebra with the physical-time evolution. The present paper specialises the Connes–Rovelli construction to a finite-dimensional Type $\mathrm{I}_{\dim_W}$ factor obtained as the wedge-restricted Connes–van Suijlekom [10] truncation of the round S^3 Camporesi–Higuchi spectral triple. The Connes–Rovelli construction is abstract; the present construction is concrete in the sense that Δ, K_{TT}, and the period closure are all explicit finite-dimensional matrix operations, computable in floating-point arithmetic at any tested $n_{\max}$.

Bisognano–Wichmann tradition and continuum modular Hamiltonians. The original Bisognano–Wichmann papers [1, 2] establish the modular-flow-as-boost identification for the Rindler wedge of Minkowski spacetime, with the analytic period $\sigma_{i\cdot 2\pi} = \mathrm{id}$ as a corollary of the Tomita–Takesaki KMS condition [26, 27] at $\beta = 2\pi$ in rapidity units. The continuum entanglement-Hamiltonian construction for a half-space in free CFT was developed by Casini and Huerta [5], and extended to spherical regions of S^d in CFT by Casini–Huerta–Myers [6]. The present construction is the operator-system-level analog of the Casini–Huerta–Myers spherical-region modular Hamiltonian on the truncated Camporesi–Higuchi triple over the round S^3, with the integer spectrum of K^W_α playing the role of the canonical 2π-period of the continuum modular flow.

Lattice realisations. Zhang *et al.* [31] verified the Bisognano–Wichmann form of the modular Hamiltonian on critical quantum spin chains, finding the continuum BW prediction to hold to good approximation on finite lattices. The present paper differs in two respects: (i) the underlying object is a continuum spectral triple on a non-flat compact S^3 background, not a lattice discretisation of physical space; (ii) the finite-dimensional structure comes from the spectral truncation of D_{CH} via the Connes–van Suijlekom framework, not from a real-space discretisation. The structural mechanism (integer-spectrum generator forcing the $\sigma_{2\pi} = \mathrm{id}$ period closure) is morally analogous to the lattice realisations of Zhang et al., but the construction lives on the operator-system side of the metric-spectral-triple framework rather than on the lattice-Hamiltonian side of condensed-matter physics.

Algebraic QFT on curved backgrounds. The standard algebraic-QFT references for modular structures on Riemannian backgrounds are Strohmaier [25] (Reeh–Schlieder on stationary spacetimes) and Verch [30] (generally covariant spin-statistics and modular structure on curved backgrounds). These address the modular-theoretic content of axiomatic QFT on curved backgrounds. The present construction is spectral-triple-internal: it does not invoke a Wightman field algebra or a nuclearity argument, but rather works directly with the $\mathcal{O}_{n_{\max}} \subset B(\mathcal{H}_{n_{\max}})$ truncation of the $C^\infty(S^3) \subset B(L^2(S^3, \Sigma))$ continuum algebra and the associated Hilbert–Schmidt GNS construction.

Lorentzian signature and Krein-space spectral triples. The proper Lorentzian extension of the present construction would require a Krein-space spectral triple at signature $(3, 1)$, for which the canonical reference is Bizi–Brouder–Besnard 2018 [3]. The Bizi–Brouder–Besnard classification organises spectral triples by pairs $(m, n) \in (\mathbb{Z}/8)^2$ where $m+n$ is the metric dimension and the underlying Connes axiom signs determine the Krein-space structure. The Riemannian Camporesi–Higuchi triple sits at $(3, 0)$; the Lorentzian extension would sit at $(3, 1)$. The Bizi–Brouder–Besnard Table 2 signs at $(3, 1)$ flip the $\{J, \gamma_5\}$ anticommutation relation, which leads to the present author's Sprint L0 prediction that the M3 sub-mechanism of the master Mellin engine trivialises at $(3, 1)$ on chirality-symmetric spectrum (`debug/lorentzian_l0_audit_memo.md` §3).

Constructing the Krein-space lift, replicating the present paper's modular construction at $(3,1)$, and verifying the M3 trivialisation prediction is the named Sprint L2 follow-up of the present author's framework documentation: a multi-month effort dominated by the absence of a published Lorentzian analog of the Latrémolière propinquity (a NCG-mathematics gap independent of any specific physical application).

Twisted-spectral-triple emergence. The recent work of Nieuviarts [21–23] proposes a twisted-spectral-triple morphism that constructs a pseudo-Riemannian spectral triple from a Riemannian almost-commutative twisted spectral triple, presented as an algebraic alternative to Wick rotation. The construction does not apply to the present S^3 case because of the structural odd-dimensional restriction (§ 3.2). The Lorentzian-extension route for the present paper therefore runs through the direct Bizi–Brouder–Besnard 2018 Krein-space lift, not through a twist morphism.

Other NCG-side gaps the present paper does not close. A genuine Lorentzian propinquity (in the sense of an analog of the Latrémolière quantum Gromov–Hausdorff propinquity for Krein-space metric spectral triples) is open as of May 2026. Toyota [28], Farsi–Latrémolière [11, 12], Rieffel [16], and Hekkelman–McDonald [15] all extend the metric-spectral-triple propinquity machinery in various directions (discrete groups of polynomial growth, collapse with abelian factors, analytic paths of Riemannian metrics, noncommutative integral on truncations, Berezin–Toeplitz on S^2), but none addresses Krein-space signature change. Constructing the Lorentzian propinquity is original NCG-mathematics work, not algorithmic.

10 Conclusion and open questions

We have established two structurally independent but operator-action conjugate constructions of the modular Hamiltonian on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$, both verifying the four-witness Wick-rotation period closure $\sigma_{2\pi}(O) = O$ *bit-exactly* at every tested $n_{\max} \in \{2,3,4,5\}$ and all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh) under the canonical hemispheric wedge P_W aligned with the Hopf-base axis, the BW local Hamiltonian $H_{\text{local}} = K_\alpha^W/\beta$ on the wedge sub-algebra, and the unit normalisation $\kappa_g = 1$ in natural rapidity units. The geometric BW-α realisation uses the integer-spectrum boost-class generator $K_\alpha^W = J_{\text{polar}}$ (Theorem 5.4); the Tomita–Takesaki BW-γ realisation uses the polar decomposition $S = J_{\text{TT}}\Delta^{1/2}$ on the GNS Hilbert–Schmidt space (Theorem 6.3). The two flows are operator-action conjugates, $\sigma_t^{\text{TT}}(O) = \sigma_{-t}^{\alpha}(O)$ (Theorem 7.1), and the four physical witnesses collapse to a single construction at the algebra-action level (Corollary 8.1).

This lifts the four-witness Wick-rotation theorem from "structural correspondence" (the published Wick-rotation chain at the metric-functional level) to *literal identification at the operator-system level* on the Riemannian truncated triple, at every finite cutoff. The structural reason is the integer spectrum of the BW local Hamiltonian, which forces $e^{i\cdot 2\pi\cdot K_\alpha^W} = I$ on the underlying Hilbert space and hence the bit-exact closure of the modular flow at $t = 2\pi$. Unlike Paper 38's qualitative-rate state-space Gromov–Hausdorff convergence (7), the present closure does not require a Gromov–Hausdorff limit and is not asymptotic in $n_{\max}$; it is a finite-cutoff exact identity.

Open questions named by the closure. (O1) Lorentzian extension at signature $(3,1)$**.** The natural next step is to replicate the present operator-system construction on a Bizi–Brouder–Besnard Krein-space spectral triple at $(3,1)$, verify the Sprint L0 prediction that M3 trivialises but M1 stays under the BBB Table 2 sign flip, and produce the genuine Lorentzian KMS-Tomita-Takesaki theorem inside the framework. The multi-month obstruction is the absence of a Lorentzian propinquity in the published literature.

Progress update (Sprints L2-B/C/D, 2026-05-16). The Hilbert-space and Lorentzian-Dirac levels of the $(3,1)$ extension have landed: Sprint L2-B (`geovac/krein_space_construction.py`) constructs the Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes L^2(\mathbb{R}_t)_{\mathrm{cutoff}}$ with fundamental symmetry $J = \gamma^0$ in the Peskin–Schroeder chiral basis, verifying all Krein axioms ($J^2 = +I$, $J^*J = I$, $J^* = J$, $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$) bit-exactly at $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$ and passing the load-bearing Riemannian-limit recovery at $N_t = 1$ bit-identically; Sprint L2-C (`geovac/lorentzian_dirac.py`) constructs $D_L = i \cdot [\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}]$ via van den Dungen Proposition 4.1 with ∂_t centered FD plus Dirichlet zero BC (structurally forced for anti-Hermiticity and Krein-self-adjointness), verifying Krein-self-adjointness $D_L^\times = D_L$ and Riemannian-limit recovery $D_L|_{N_t=1} = iD_{\mathrm{GV}}$ bit-exactly across the same panel. Sprint L2-D (`geovac/connes_axiom_audit_31.py`) verifies four of the six BBB-predicted axioms at $(m,n) = (4,6)$ bit-exactly ($J_L^2 = +I$, $\{J_L, \gamma^5\} = 0$, $\{J_L, \eta\} = 0$, $J_L D_L = +D_L J_L$); the fifth axiom $\{\gamma^5, D_L\} = 0$ is a *structural finding*, failing on the truthful Camporesi–Higuchi Dirac (which is chirality-diagonal in the chiral basis and therefore commutes with γ^5) and requiring an R1+R2 split documented in Paper 32 § IV / § VIII.D [35]. This is structurally analogous to the Riemannian-side R3.5/present-paper trade between truthful CH (correct for axioms requiring $JD = +DJ$) and offdiag CH (correct for SDP-bounding Connes-distance constructions). The construction does *not* re-open WH1 PROVEN (Paper 38 [37]); the Krein construction is an extension on top of the Riemannian foundation, not a re-test of it. Full operator-system-level closure (the analog of the present paper's literal identification of the four-witness theorem at Lorentzian signature) is the named Sprint L2-E target, still open at the multi-month scale due to the absence of a published Lorentzian propinquity. O1 status remains OPEN *at the continuum / non-compact level*; the Hilbert-space and Dirac-operator levels are now closed at finite cutoff. The *metric*-level finite-cutoff question is now understood structurally (Paper 45 § open Q1, 2026-06): the truncated Bisognano–Wichmann boost is compact (integer spectrum, $e^{2\pi i K} = I$), so the finite-cutoff quantum metric is Euclidean and the Lorentzian signature is the Wick rotation of the KMS circle — a convention, with the genuine reverse-triangle signature confined to the non-compact continuum limit.

(O2) Non-tracial wedge states. The unified closure rests on the tracial-Gibbs structure $\rho_W = e^{-K_\alpha^W}/Z$ on a finite-dimensional Type $\mathrm{I}_{\dim_W}$ factor. For non-tracial wedge states (e.g., excited states above the BW vacuum, mixed states with non-Gibbs structure), the polar decomposition $S = J_{\mathrm{TT}}\Delta^{1/2}$ still goes through, but the integer-spectrum property of K_{TT} is not guaranteed, and the bit-exact closure may degrade to a soft propinquity-rate-controlled closure of the type Paper 38 supplies. The structural status of the non-tracial case is open.

(O3) Spectral-action Dirac vs. modular-Hamiltonian generator. Section 7.2 flagged that the framework's intrinsic Camporesi–Higuchi Dirac D_{CH} is *not* the right local Hamiltonian for the BW vacuum at $\beta = 2\pi$: the BW vacuum selects K_α^W, not D_{CH}, as its modular generator. On the round S^3 this distinction is mostly a matter of choosing the right generator from a family of admissible candidates, but in general the spectral-action D_{CH} and the modular-Hamiltonian K_α^W may not even commute. The structural status of this distinction in extensions to other compact-Lie-group spectral triples (per Paper 40 [39]) is open and an interesting direction for follow-up.

Refinement (Sprint L2-E, 2026-05-16; § 11 below). The O3 question of signature-dependence is now *partially* closed: the $H_{\mathrm{local}} \neq D_W$ distinction is **signature-INDEPENDENT at the Riemannian limit** and refined upward at $N_t > 1$ by temporal-derivative content. At $N_t = 1$ on the Lorentzian Krein space, $\|H_{\mathrm{local}} - D_L^W\|_F$ equals the Riemannian-side residual $\|H_{\mathrm{local}} - D_{\mathrm{GV}}^W\|_F$ *bit-exact* at every tested $n_{\max} \in \{1,2,3\}$. The distinction is therefore NOT a peculiarity of the round-S^3 Riemannian truncation but a deeper structural feature of the framework's modular content. At $N_t > 1$ the Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I)$ has temporal-derivative content that K_α^W/β cannot capture; the gap between the spectral-action and modular-Hamiltonian generators widens with temporal cutoff. The *deeper* question (why is the spectral-action Dirac not the modular generator on the BW vacuum, structurally?) remains open and is the residual O3.

Further refinement (Pythagorean orthogonality, 2026-05-17; Remark 8.2 above, Paper 43 §10.2 [40]). A post-L2-E PSLQ probe has identified the structural mechanism behind the signature-independent residual: H_{local} and D_W^L are Hilbert–Schmidt orthogonal, $\langle H_{\text{local}}, D_W^L\rangle_{\text{HS}} = 0$ bit-exact across the entire 18-cell six-witness panel, and the squared residual decomposes as a Pythagorean sum $\|H_{\text{local}} - D_W^L\|_F^2 = \kappa_g^2 S(n_{\max})/(4\pi^2) + D(n_{\max})$ with S, D pure rationals in $n_{\max}$. The sole transcendental $(1/\pi^2)$ places the residual entirely in the Paper 32 § VIII master Mellin engine M1 sector. The mechanism is the diagonal/off-diagonal subspace decomposition of $\mathcal{B}(\mathcal{H}_W)$: H_{local} is diagonal in the unfolded wedge spinor basis, D_W^L is off-diagonal. The S^3-specificity of this construction — and its failure on the Bargmann–Segal S^5 Hardy sector, which lacks a spinor sector and a half-integer wedge — is the fourth layer of the Paper 24 § 5.3 [33] Coulomb/HO asymmetry (alongside spectrum-computing L_0, calibration π, and Wilson matter coupling). The structural-explanation question (*why* is D_W the wrong modular generator?) is now sharpened to a subspace-orthogonality statement; a formal proof of the diagonal/off-diagonal decomposition at general $n_{\max}$ is a named follow-on.

(O4) Higher-rank compact non-abelian extensions. Paper 40 [39] extends the Paper 38 state-space Gromov–Hausdorff convergence theorem to all compact connected Lie groups with bi-invariant metric. The natural extension of the present paper would verify that the geometric BW-α generator analog on higher-rank groups (e.g., $K_\alpha^W = J_z$ with J_z the Cartan generator along the wedge-defining direction) still has integer spectrum on the spinor harmonic basis, preserving the bit-exact period closure across the universal $4/\pi$-rate class. The standard structure of weights in compact Lie groups suggests this extension is straightforward; verifying it explicitly is a near-term sprint follow-up.

(O5) Cross-manifold modular structures. The Paper 24 [33] cross-manifold blocker $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ remains unaddressed by the present paper. A cross-manifold modular construction (modular Hamiltonian on a tensor product of a round S^3 Camporesi–Higuchi triple and a holomorphic-sector S^5 Bargmann–Segal triple) would require an analog of the Latrémolière tensor-product propinquity of Paper 39 [38] extended to mixed Riemannian / Hardy-sector spectral triples. This is also open.

Final remark on the M1 master Mellin engine identification. The 2π in the modular period of the present construction is the same 2π that drives the Paper 38 state-space-GH rate $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ via the M1 Hopf-base measure mechanism (Paper 32 § VIII case-exhaustion theorem). The present paper establishes this identification at the operator-system level, not just at the metric-functional level: the modular period $\sigma_{2\pi}(O) = O$ is a framework-internal output of the spectral truncation, with the integer-spectrum generator K_α^W acting as the mechanism through which the M1 transcendental signature $\mathrm{Vol}(S^1) = 2\pi$ is realised at the algebra-action level. The framework's M1 mechanism on the round S^3 Camporesi–Higuchi triple is now operator-system-level identified with the four-witness Wick-rotation period structure, in both the geometric and the Tomita–Takesaki readings, at every finite cutoff.

11 Lorentzian closure at finite cutoff (Sprint L2)

The body of this paper closes the four-witness Wick-rotation theorem at the operator-system level on the Riemannian truncated triple $\mathcal{T}_{n_{\max}}$. Open question (O1) of §10 named the Lorentzian extension to signature $(3, 1)$ as the next step. Sprint L2 (May 2026) closes that extension at finite cutoff via five sub-sprints (L2-A scoping, L2-B Krein space, L2-C Lorentzian Dirac, L2-D Connes axiom audit, L2-E modular Hamiltonian). This section documents the closure and the substantive structural finding it produced; the math.OA-style standalone writeup is outlined in Paper 43 (`papers/group1_operator_algebras/paper_43_lorentzian_extension_outline.md`, companion).

11.1 The five-sub-sprint architecture

The Sprint L2 architecture mirrors the Riemannian construction of §§2–8 verbatim with a temporal slot added. At signature $(3, 1)$ West-coast in the Peskin–Schroeder chiral basis (Sprint L2-B convention lock-in):

- **Krein space (L2-B).** $\mathcal{K}_{n_{\max},N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$, with N_t a bounded uniform temporal grid on $[-T_{\max}, T_{\max}]$ (NOT compactified to S^1_β, which would re-introduce closed timelike curves per Geroch's theorem). Fundamental symmetry $J = \gamma^0 \otimes I_{N_t}$ with γ^0 the chirality-swap in the chiral basis. All Krein axioms ($J^2 = +I$, $J^*J = I$, $J^* = J$, $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ orthogonally with $\dim \mathcal{K}^\pm = \dim \mathcal{K}/2$) are bit-exact (Frobenius residual = 0.0 in float64) at every $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$.
- **Lorentzian Dirac (L2-C).** $D_L := i \cdot \left[\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\right]$ per van den Dungen Proposition 4.1 (arXiv:1505.01939) with ∂_t centered finite-difference plus Dirichlet zero BC (structurally forced for anti-Hermiticity). Krein-self-adjointness $D_L^\times = D_L$ is bit-exact across the panel; the sign $i^t = +i$ for $t = 1$ is derived from the Krein-self-adjointness requirement, not chosen.
- **Connes axiom audit (L2-D).** At Bizi–Brouder–Besnard [3] signature $(m, n) = (4, 6)$ (the BBB Table 3 translation of Lorentzian $(s, t) = (3, 1)$ West-coast), four BBB-predicted-sign axioms hold bit-exactly: $J_L^2 = +I$ ($\varepsilon = +1$); $\{J_L, \gamma^5\} = 0$ ($\varepsilon'' = -1$); $\{J_L, \eta\} = 0$ ($\varepsilon\kappa = -1$); and the BBB universal $J_L D_L = +D_L J_L$. See Paper 32 § VIII.D [35] for the full audit.
- **Modular Hamiltonian (L2-E, this closure).** Built on the wedge $W_L := P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$, with P_W^{spatial} the Definition 4.1 hemispheric wedge on $\mathcal{H}_{\mathrm{GV}}$ and $P_{t\geq 0}$ the $t \geq 0$ half-line projector. Wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ under the BW choice $H_{\mathrm{local}} := K_L^{\alpha,W}/\beta$ with $\beta = 2\pi$. The construction mirrors §§5–6 of the Riemannian closure, with the temporal slot adding additional content at $N_t > 1$.

The architecture is structurally additive on top of the Riemannian closure of this paper: at $N_t = 1$, every Sprint L2 construction reduces *bit-identically* to the corresponding Riemannian-side object of §§2–8.

11.2 The Krein-level four-witness theorem at finite cutoff

Theorem 11.1 (Krein-level four-witness Wick-rotation theorem at finite cutoff)**.** *On the Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t}$ of §11.1, with the wedge $W_L = P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$, the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ under the BW choice $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$, and the unit normalisation $\kappa_g = 1$ in natural rapidity units, all of the following hold at every tested $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$:*

(a) BW-α period closure (geometric). *For every $O \in B(\mathcal{K}_{W_L})$, $\sigma_{2\pi}^{L,\alpha}(O) = O$ bit-exactly, with periodicity residual $\leq 4 \times 10^{-16}$ (machine precision).*

(b) BW-γ Tomita period closure. *On the Krein-GNS Hilbert–Schmidt space $\mathcal{K}_{\mathrm{GNS}} = M_{\dim W_L}(\mathbb{C})$ with modular operator Δ_L and modular Hamiltonian $K_L^{\mathrm{TT}} = -\log \Delta_L$, $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$ bit-exactly at the same $\leq 4 \times 10^{-16}$ residual.*

(c) Flow conjugacy at general t. *$\sigma_t^{L,\mathrm{TT}}(a) = \sigma_{-t}^{L,\alpha}(a)$ bit-exactly for $t \in \{1, \pi, 2\pi\}$ on five sampled multipliers, residual $\leq 4 \times 10^{-16}$.*

(d) Six-witness collapse. *For all six instantiations $\{$BW, HH$_{M=1}$, HH$_{M=2}$, SEW$_{M=1}$, UNRUH$_{a=1}$, UNRUH$_{a=2}\}$, the cross-witness consistency residual is exactly 0.0.*

Proof sketch. $K_L^{\alpha,W}$ is diagonal with integer eigenvalues two_m_j (inherited from Definition 5.1 on the spatial side, with the temporal slot acting as I_{N_t}), so $e^{i\cdot 2\pi\cdot K_L^{\alpha,W}} = I$ identically — proving (a). Statement (b) follows from $\sigma_t^{L,\mathrm{TT}}(a) = e^{-itK_L^{\alpha,W}}\, a\, e^{+itK_L^{\alpha,W}}$ (polar decomposition + $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ identity) combined with the integer-spectrum property of $K_L^{\alpha,W}$. Statement (c) is the same operator identity at general t. Statement (d) follows from the β-independence of ρ_W^L under the BW choice $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$: the witnesses parameterise different physical κ_g, but the modular construction does not depend on κ_g at the algebra-action level. Full proofs mirror Theorems 5.4, 6.3, 7.1, and Corollary 8.1 of the Riemannian closure; the numerical verification at every panel cell is documented in `debug/data/l2_e_modular_hamiltonian_lorentzian.json` via `geovac/modular_hamiltonian_lorentzian.py`. □

The headline reading: the four-witness Wick-rotation theorem (Hartle–Hawking + Sewell + Bisognano–Wichmann + Unruh) is lifted from *structural correspondence at the metric-functional level* (Sprint TD Track 4 + Sprint Unruh-pendant) to *literal identification at the operator-system level (Lorentzian, finite cutoff)* at every tested Krein cutoff. This is the framework's first operator-system-level literal identification of a Lorentzian QFT theorem at finite cutoff.

11.3 Riemannian-limit identification (bit-identical at $N_t = 1$)

The Sprint L2 architecture is the genuine Lorentzian *extension* of the Riemannian closure of §§2–8, not a parallel-but-different construction:

Proposition 11.2 (Riemannian-limit recovery at $N_t = 1$)**.** *At $N_t = 1$, every Sprint L2 construction reduces bit-identically (Frobenius residual exactly* 0.0*, not merely machine-precision) to the corresponding Riemannian-side object of this paper:*

- $P_{W_L}|_{N_t=1} = P_W^{\mathrm{spatial}}$ *(Definition 4.1);*
- $K_L^{\alpha}|_{N_t=1} = K_\alpha$ *(Definition 5.1);*
- $K_L^{\alpha,W}|_{N_t=1} = K_\alpha^W$*;*
- $\rho_W^L|_{N_t=1} = \rho_W$ *(§4.2);*
- $\Delta_L|_{N_t=1} = \Delta$ *(§6);*
- $K_L^{\mathrm{TT}}|_{N_t=1} = K_{\mathrm{TT}}$ *(§6).*

The bit-identical reduction is structural: at $N_t = 1$ the temporal-derivative piece ∂_t vanishes on the singleton grid, and the Kronecker product with $I_1 = 1$ is the identity operation. Every Riemannian-side bit-exact property (Theorems 5.4, 6.3, 7.1) is preserved verbatim at $N_t = 1$.

Verification. Bit-identical at $n_{\max} \in \{1, 2, 3\}$, documented in `tests/test_modular_hamiltonian_lorentzi`

This Riemannian-limit identification is the load-bearing falsifier that ensures Sprint L2-E is the Lorentzian extension of the present paper, not a parallel construction. The temporal slot adds genuine new content at $N_t > 1$ (the temporal-derivative refinement of §11.4 below) that is invisible at $N_t = 1$.

11.4 The H_{local} verdict at $(3, 1)$ — signature-independent at Riemannian limit

This subsection states the substantive structural finding of Sprint L2-E. Open question O3 of §10 (the spectral-action vs. modular-Hamiltonian generator distinction flagged in §7.2 as the "derived-Hamiltonian finding") is refined at finite cutoff:

Theorem 11.3 (H_{local} signature-independence at Riemannian limit)**.** *At $N_t = 1$ (Riemannian limit on the Krein space at $(3,1)$),*

$$\left\| H_{\text{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1} \right\|_F \;=\; \left\| H_{\text{local}}|_{(3,0)} - D_{\text{GV}}^W \right\|_F$$

bit-exact at every tested $n_{\max} \in \{1,2,3\}$: values 2.1332, 6.5275, 13.854 *respectively, matching the Riemannian residual of the present paper at the same $n_{\max}$. At $N_t > 1$, the Lorentzian residual is refined upward by the temporal-derivative contribution to D_L^W, growing as $O(\sqrt{N_t})$ but with the Riemannian baseline preserved.*

Mechanism. At $N_t = 1$, $D_L = i \cdot D_{\text{GV}}$ exactly (the temporal derivative ∂_t vanishes on the singleton grid). Wedge restriction gives $D_L^W = i \cdot D_{\text{GV}}^W$. The norm $\|H_{\text{local}} - i \cdot D_{\text{GV}}^W\|_F$ equals $\|H_{\text{local}} - D_{\text{GV}}^W\|_F$ when H_{local} is real (which it is — $K_L^{\alpha,W}/\beta$ has real integer-rational diagonal entries). So the residuals are bit-identical.

Structural reading. The §7.2 / O3 distinction between the spectral-action Dirac D_W and the modular-Hamiltonian generator K_α^W/β on the BW vacuum is **not a peculiarity of the round-S^3 Riemannian truncation**. It persists bit-exactly at signature $(3,1)$ at the Riemannian limit, and is refined upward at $N_t > 1$ by the temporal-derivative content of D_L. The distinction is therefore a **deeper structural feature** of the framework's modular content, independent of signature at the Riemannian-limit baseline.

The temporal-derivative refinement at $N_t > 1$ is a Lorentzian-side enrichment of the Riemannian finding: at $(3,1)$, D_L has more content than D_{GV} (it includes the temporal derivative), so the gap between the spectral-action object and the modular-Hamiltonian generator widens with temporal cutoff. This is not a sign that §7.2 was wrong; it is a sign that the Lorentzian extension reveals *additional* structure in the gap.

This finding refines Paper 42 O3 from "open question about signature-dependence" to "signature-INDEPENDENT structural distinction at the Riemannian limit, refined by temporal-derivative content at $N_t > 1$." The *deeper* question (why is the spectral-action Dirac not the modular generator on the BW vacuum, structurally?) remains the residual O3 open question.

11.5 Honest scope at finite cutoff

The closure of Theorem 11.1 is at finite $n_{\max}$ and finite N_t — **not** in the continuum (GH-limit, Lorentzian propinquity) limit. Explicitly:

- The Latrémolière propinquity [17, 18] used in Papers 38–40 for the Riemannian-side GH-convergence theorem is purely Riemannian as of May 2026. Subsequent literature (Toyota [28], Farsi–Latrémolière 2024/2025 [11, 12], Hekkelman–McDonald 2024 a/b [15]) covers only Riemannian / Hilbert-space structures. No published Lorentzian propinquity construction exists.
- Constructing a Lorentzian propinquity is the named Sprint L3 target (6–12 months of original NCG mathematics). The Nieuviarts 2025 [22, 23] twist-morphism may shorten this budget if the construction applies to GeoVac $S^3 = \text{SU}(2)$ (odd-dimension caveat that Sprint L0 audit flagged for a separate L2-Nieuviarts-scoping pass).
- The Sprint L2-D structural finding (BBB universal axiom $\chi D = -D\chi$ fails on truthful Camporesi–Higuchi D_{GV}) is the load-bearing scope finding: the Sprint L2-E construction uses truthful D_{GV} (which preserves the Riemannian-limit recovery of Proposition 11.2), not a full BBB indefinite spectral triple in the strict Sec 5(v) sense. The wedge-modular construction is $K_L^{\alpha,W}$-driven, not D_L-driven, so the period closure is D_L-independent at the operator-action level. Paper 32 § VIII.D [35] documents the three resolutions (R1 accept, R2 use offdiag CH, R3 redefine γ^5); Theorem 11.1 uses R1 (truthful D_{GV}).

The verdict **closed-at-finite-cutoff** is therefore an honest finite-cutoff statement. Theorem 11.1 is the framework's first operator-system literal identification of a Lorentzian QFT theorem; the Lorentzian-propinquity rate-level extension is the named open Sprint L3 target.

11.6 References and pointer to standalone writeup

The construction recipe is van den Dungen 2016 Proposition 4.1 (arXiv:1505.01939): Wick-rotate $g \to g_r$ (positive-definite), construct the standard Riemannian Dirac $\not{D}_{g_r}$, and set $D_L = i^t \cdot \not{D}_{g_r}$ as the Krein-self-adjoint Lorentzian Dirac with $\mathcal{J} = \gamma^0$. At $(s,t) = (3,1)$ this gives $i^t = i$ and the construction of §11.1.

The Bizi–Brouder–Besnard 2018 $(m,n) \in (\mathbb{Z}/8)^2$ classification [3] provides the signature-dependent Connes axiom signs: the $(s,t) = (3,1)$ West-coast entry sits at $(m,n) = (4,6)$ via BBB Table 3, verified directly from the BBB PDF in Sprint L2-D (Paper 32 § VIII.D).

The full math.OA-style writeup of the Lorentzian extension is outlined in `papers/group1_operator_algebra` (companion document). When drafted as a Zenodo deposit, Paper 43 should **not supersede** the present paper: Paper 42 is the Riemannian closure of the four-witness theorem, Paper 43 is the Lorentzian extension, and together they constitute the four-witness Wick-rotation arc at the operator-system level.

The sprint memos in `debug/l2_{a,b,c,d,e,paper_edits_round1}_*.md` and the synthesis memo `debug/sprint_l2_synthesis_memo.md` document the full Sprint L2 architecture and the substantive structural findings; the computational drivers and verification data sit in `geovac/{krein_space_con` `lorentzian_dirac.py, connes_axiom_audit_31.py, modular_hamiltonian_lorentzian.py}` and the corresponding `tests/test_*.py` files.

A Computational verification protocol

The numerical verification of the period closures of Theorems 5.4 and 6.3 and of the cross-witness collapse of Table 3 is implemented in the supporting Python module `geovac/modular_hamiltonian.py` of the present author's framework codebase (∼1550 lines), with test coverage in `tests/test_modular_hamiltoni` (∼960 lines, 67 tests: 63 fast plus 4 slow). The structure is briefly described here.

A.1 Module API

The principal classes are:

- `HemisphericWedge(axis="hopf")`: implements the projection P_W of Definition 4.1 via the equatorial-reflection involution R_{polar} on the full-Dirac spinor basis.
- `TomitaConjugation(U)`: implements the antilinear J_{TT} via a stored unitary U with action $J_{\mathrm{TT}}(\psi) = U \cdot \overline{\psi}$; exposes `verify_J_squared_positive_identity` and `verify_J_squared_negative_iden` as explicit sanity checks distinguishing J_{TT} from J_{GV}.
- `KMSState(beta, H)`: implements the wedge-restricted Gibbs state $\omega_{\beta,H}$ of (10), including the canonical analytic-continuation verification of the KMS condition at the expectation level.
- `ModularHamiltonian(wedge, basis, D, kappa_g)`: implements the BW-α construction with $K = K_\alpha^W$, the real-time flow σ_t^α, the period-closure verifier `verify_modular_periodicity`, and accessors to the Tomita–Takesaki structure (`tomita_structure`, `k_alpha_geometric`, `k_tomita`, `compare_alpha_vs_tomita`).
- `TomitaModularStructure(rho)`: implements the BW-γ Tomita construction on the GNS Hilbert–Schmidt space, with $\Delta = (\rho^{-1})^T \otimes \rho$, $K_{\mathrm{TT}} = -\log \Delta$, and the modular flow $\sigma_t^{\mathrm{TT}}(a) = \rho^{it} a \rho^{-it}$ on the algebra.

Witness factories for_bisognano_wichmann, for_hartle_hawking, for_sewell, for_unruh instantiate ModularHamiltonian with witness-specific κ_g values: BW canonical ($\kappa_g = 1$), HH ($\kappa_g = 1/(4M)$), Unruh ($\kappa_g = a$). Each factory returns a ModularHamiltonian object whose period-closure verifier returns STRONG_IDENTIFICATION at every tested $n_{\max}$.

A.2 Test coverage

The test suite tests/test_modular_hamiltonian.py verifies the following: wedge projector idempotency and Hermiticity ($P_W^2 = P_W$, $P_W^* = P_W$); Tomita conjugation properties ($J_{\mathrm{TT}}^2 = +I$, $J_{\mathrm{TT}}^2 \neq -I$, antilinearity, operator-action consistency); KMS state normalisation, partition-function positivity, real expectation on Hermitian operators; modular periodicity $\sigma_{2\pi}(O) = O$ at $n_{\max} \in \{2, 3, 4\}$ for the first five non-identity multipliers in FullDiracTruncatedOperatorSystem; cross-witness collapse at the operator-action level; KMS expectation-level condition for sample operator pairs (A, B); propinquity rate cross-check (residual $\ll \gamma_{n_{\max}}$); operator-system non-leakage ($\sigma_{2\pi}(M)$ stays in $\mathcal{O}_{n_{\max}}$); modular flow group law $\sigma_t \cdot \sigma_s = \sigma_{t+s}$ and identity at $t = 0$. The slow tests verify the closure at $n_{\max} = 5$ ($\sim$80 seconds wall time per slow test).

A.3 Wall time and reproducibility

Per-$n_{\max}$ wall time on a standard laptop (Intel i7, Python 3.11, NumPy 1.26): $\sim$ 0.4 s at $n_{\max} = 2$, $\sim$ 3.5 s at $n_{\max} = 3$, $\sim$ 45 s at $n_{\max} = 4$, $\sim$ 260 s at $n_{\max} = 5$ for the BW-α construction; $\sim$ 0.2 s, $\sim$ 3.7 s, $\sim$ 45 s, $\sim$ 545 s for the BW-γ construction (the $n_{\max} = 5$ cost dominated by the 4900×4900 GNS-Hilbert–Schmidt diagonalisation). Beyond $n_{\max} = 5$ the GNS construction becomes expensive; the structural reading of the closure is unchanged at higher $n_{\max}$ since the integer-spectrum mechanism holds at any cutoff. Computational drivers and structured output JSON are available in the present author's framework documentation under debug/l1_modular_hamiltonian_compute.py and debug/l1_tighten_tomita_compute.py respectively.

B Cross-references to the GeoVac framework

The present paper is part of an integrated NCG-research programme developed by the present author. We briefly note the principal cross-references for the reader's orientation.

Paper 38 ([37], state-space GH convergence on S^3). The convergence of the truncated Camporesi–Higuchi spectral triples $\mathcal{T}_{n_{\max}}$ to the continuum $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance is established in Paper 38 with explicit rate (7). The L1' / L2 / L3 / L4 / L5 lemmas of Paper 38 supply the metric-side analytic infrastructure that the present paper builds on. The state-space GH rate constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ identifies as the same Hopf-base measure factor that drives the modular period closure of the present Theorems 5.4–6.3.

Paper 40 ([39], universal $4/\pi$ on compact Lie groups). The Paper 38 state-space Gromov–Hausdorff convergence extends to all compact connected Lie groups with bi-invariant metric, with the rate constant $4/\pi$ universal across the class via the Plancherel-weight $\times$ Vandermonde-Jacobian cancellation theorem (Paper 40 § 3.2). The modular structure of the present paper plausibly extends to the higher-rank cases (Section 10 O4); the integer spectrum of the boost-class generator K_α^W on the spinor harmonic basis is a property of SU(2) (where the half-integer weights double to integers), and the higher-rank analog requires verifying that the relevant Cartan generator along the wedge axis still has the right spectral property.

Paper 32 §VIII (master Mellin engine case-exhaustion theorem). The case-exhaustion theorem [35, Thm. thm:pi_source_case_exhaustion] of Paper 32 §VIII (in the present author's framework documentation) states that every π in any finite chain of Paper 34 projections engages one of three master Mellin engine sub-mechanisms: M1 (Hopf-base measure $\mathrm{Vol}(S^2)/\pi^2$ at operator order $k = 0$), M2 (Seeley–DeWitt $\sqrt{\pi}\,\mathbb{Q}$ at $k = 2$), or M3 (vertex-parity Hurwitz / Catalan G / Dirichlet L at $k = 1$). The 2π in the modular period of the present paper is the M1 signature at the operator-system level. The Bisognano–Wichmann reading of M1 on temporal compactifications ([35, Rem. rem:bisognano_wichmann_reading]) is exactly the structural correspondence that the present paper lifts to literal identification.

Paper 34 §III.27 (Wick rotation projection). The Wick rotation projection in the present author's projection taxonomy [36] is the structural mechanism through which the framework's Euclidean spectral content is read as a Lorentzian physical observable. The Sprint L0 audit (`debug/lorentzian_l0_audit_memo.md`) classifies this projection in the WICK_MAP_FREE bucket at the metric-functional level. The present paper extends the projection's status to literal identification at the operator-system level (Riemannian), reducing the framework's reliance on the published Wick-rotation chain for the period-closure identity itself.

Paper 24 ([33], Coulomb / HO asymmetry). The cross-manifold blocker named in Paper 24 §V (Riemannian $S^3 \otimes$ holomorphic-sector S^5 Bargmann–Segal triple) carries through to the modular construction: the present paper does not address the cross-manifold modular structure, and the natural extension would require a mixed Riemannian / Hardy-sector tensor-product propinquity that does not yet exist in the literature.

Paper 2 ([32], fine-structure-constant observation). The present paper does not impinge on Paper 2's status as an Observation. The $n_{\max} = 3$ cutoff of Table 1 happens to coincide with Paper 2's selection-principle truncation ($\dim \mathcal{H}_3 = 40 = g_3^{\mathrm{Dirac}} = \Delta^{-1}$); this is a structural coincidence at the Hilbert-space dimension level, not a derivation of Paper 2's combination rule $K = \pi(B + F - \Delta)$. The combination rule remains a numerical observation without first-principles derivation.

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